

Linear (Gaussian) Models

Contents

Configuring R	1
1 Overview	2
1.1 Why this course includes linear regression	2
1.2 Chapter overview	2
2 Understanding Gaussian Linear Regression Models	3
2.1 Motivating example: birthweights and gestational age	3
2.2 Dobson <code>birthweight</code> data	3
2.2.1 Data as table	3
2.2.2 Reshape data for graphing	3
2.2.3 Data as graph	3
2.2.4 Data notation	5
2.3 Parallel lines regression	6
2.3.1 Model assumptions and predictions	7
2.3.2 Coefficient Interpretation	9
2.4 Interactions	11
2.4.1 Coefficient Interpretation	14
2.4.2 Compare coefficient interpretations	15
2.5 Stratified regression	15
2.6 Curved-line regression	16
2.7 Rescaling	18
2.7.1 Rescale age	18
3 Estimating Linear Models via Maximum Likelihood	20
3.0.1 Likelihood	21
3.0.2 Log-likelihood	21
3.0.3 Score function	22
3.0.4 Hessian	23
3.0.5 Alternative approach using matrix derivatives	24
3.0.6 Residual Standard Deviation	26
4 Inference about Gaussian Linear Regression Models	26
4.1 Motivating example: <code>birthweight</code> data	26
4.2 Wald tests and CIs	27
4.3 P-values	27
4.4 Confidence intervals	27
4.5 Gaussian approximations	28
4.6 P-values	28
4.7 Confidence intervals	28
4.8 Likelihood ratio statistics	29
5 Goodness of fit	30
5.1 AIC and BIC	30
5.1.1 AIC in R	31
5.1.2 BIC in R	31

5.2	(Residual) Deviance	31
5.2.1	Saturated models	32
5.3	Null Deviance	34
6	Prediction	36
6.1	Prediction for linear models	36
6.2	Example: prediction for the <code>birthweight</code> data	36
6.3	Confidence intervals	37
6.4	Prediction intervals	38
7	Diagnostics	40
7.1	Assumptions in linear regression models	40
7.2	Direct visualization	41
7.2.1	Fitted model for <code>hers</code> data	41
7.3	Residuals	43
7.3.1	Residuals of fitted values vs residual noise	45
7.3.2	General characteristics of residuals	45
7.3.3	Characteristics of residuals in Gaussian models	46
7.3.4	Computing residuals in R	46
7.3.5	Graphing the residuals	47
7.3.6	Residuals versus predictors	48
7.3.7	Residuals versus fitted values	48
7.4	Marginal distributions of residuals	51
7.4.1	Standardized residuals in R	51
7.5	QQ plot of standardized residuals	54
7.5.1	QQ plot - how it's built	55
7.6	Conditional distributions of residuals	56
7.6.1	Residuals versus fitted values	57
7.6.2	Scale-location plot	62
7.6.3	Residuals versus leverage	63
7.7	Diagnostics constructed by hand	64
8	Model selection	67
8.1	Mean squared error	67
8.1.1	Cross-validation	67
8.1.2	stepwise regression	69
8.1.3	Lasso	70
9	Categorical covariates with more than two levels	72
9.1	Example: <code>birthweight</code>	72
9.2		72
9.3	Let's see how that model looks:	74
9.4	Let's see what R does with categorical variables by default:	74
9.5	Re-parametrize with no intercept	75
9.6	Let's see what these new models look like:	75
9.7	Let's see how R did that:	76
10	Ordinal covariates	77

Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
```

```

library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times

```

Here are some R settings I use in this document:

```

rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE

include_reference_lines <- FALSE

```

Note

This content is adapted from:

- Dobson and Barnett (2018), Chapters 2-6
- Dunn and Smyth (2018), Chapters 2-3
- Vittinghoff et al. (2012), Chapter 4

There are numerous textbooks specifically for linear regression, including:

- Kutner et al. (2005): used for UCLA Biostatistics MS level linear models class
- Chatterjee and Hadi (2015): used for Stanford MS-level linear models class
- Seber and Lee (2012): used for UCLA Biostatistics PhD level linear models class and

- UC Davis STA 108.
- Kleinbaum et al. (2014): same first author as Kleinbaum and Klein (2010) and Kleinbaum and Klein (2012)
- *Linear Models with R* (Faraway 2025)
- *Applied Linear Regression* by Sanford Weisberg (Weisberg 2005)

For more recommendations, see the discussion on Reddit^a.

- see also <https://web.stanford.edu/class/stats191>¹

^ahttps://www.reddit.com/r/statistics/comments/qwgctl/q_books_on_applied_linear_modelsregression_for/

1 Overview

1.1 Why this course includes linear regression

- This course is about *generalized linear models* (for non-Gaussian outcomes)
- UC Davis STA 108 (“Applied Statistical Methods: Regression Analysis”) is a prerequisite for this course, so everyone here should have some understanding of linear regression already.
- We will review linear regression to:
- make sure everyone is caught up
- provide an epidemiological perspective on model interpretation.

1.2 Chapter overview

- Section 2: how to interpret linear regression models
- Section 3: how to estimate linear regression models
- Section 4: how to quantify uncertainty about our estimates
- Section 7: how to tell if your model is insufficiently complex

2 Understanding Gaussian Linear Regression Models

2.1 Motivating example: birthweights and gestational age

Suppose we want to learn about the distributions of birthweights (*outcome* Y) for (human) babies born at different gestational ages (*covariate* A) and with different chromosomal sexes (*covariate* S) (Dobson and Barnett (2018) Example 2.2.2).

2.2 Dobson birthweight data

2.2.1 Data as table

2.2.2 Reshape data for graphing

2.2.3 Data as graph

```
plot1 <- bw |>
  ggplot(aes(
    x = age,
    y = weight,
    shape = sex,
    col = sex
  )) +
```

¹the current version of the first regression course I ever took

Table 1: `birthweight` data (Dobson and Barnett (2018) Example 2.2.2)

```
library(dobson)
data("birthweight", package = "dobson")
birthweight
#> # A tibble: 12 x 4
#>   `boys gestational age` `boys weight` `girls gestational age` `girls weight`
#>   <dbl> <dbl> <dbl> <dbl>
#> 1      40      2968      40      3317
#> 2      38      2795      36      2729
#> 3      40      3163      40      2935
#> 4      35      2925      38      2754
#> 5      36      2625      42      3210
#> 6      37      2847      39      2817
#> 7      41      3292      40      3126
#> 8      40      3473      37      2539
#> 9      37      2628      36      2412
#> 10     38      3176      38      2991
#> 11     40      3421      39      2875
#> 12     38      2975      40      3231
```

```
theme_bw() +
  xlab("Gestational age (weeks)") +
  ylab("Birthweight (grams)") +
  theme(legend.position = "bottom") +
  # expand_limits(y = 0, x = 0) +
  geom_point(alpha = .7)
print(plot1 + facet_wrap(~sex))
```

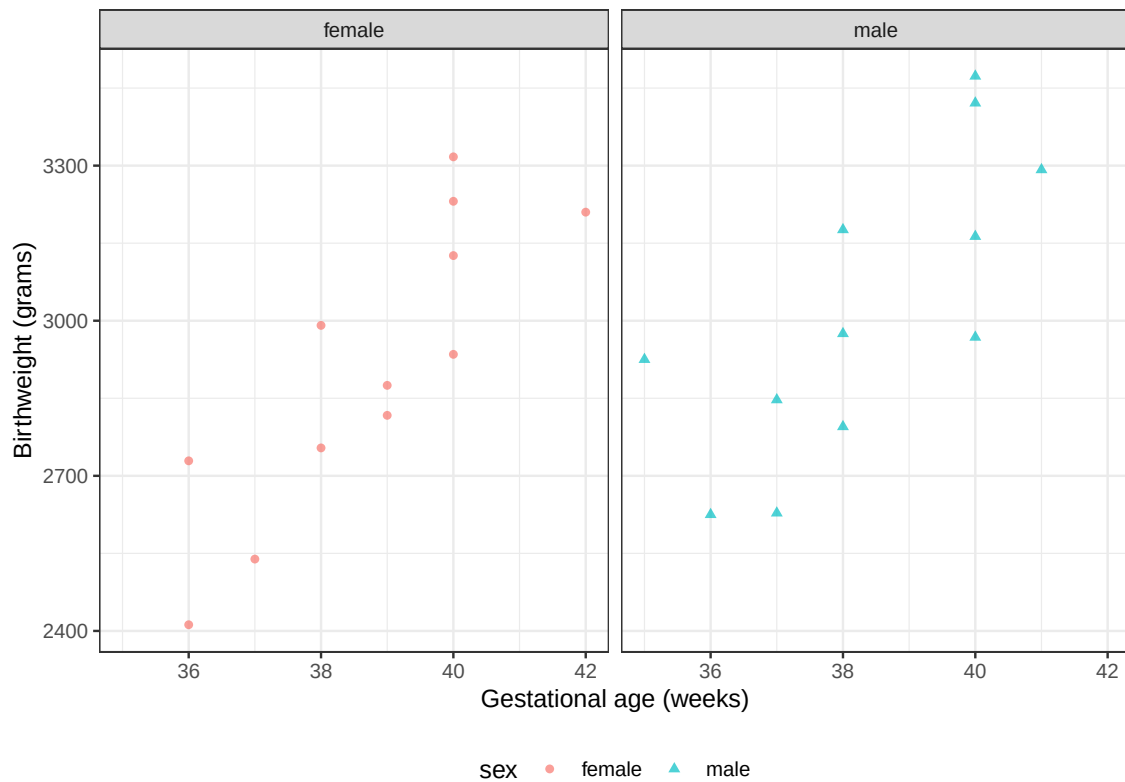


Figure 1: `birthweight` data (Dobson and Barnett (2018) Example 2.2.2)

Table 2: `birthweight` data reshaped

```
library(tidyverse)
bw <-
  birthweight |>
  pivot_longer(
    cols = everything(),
    names_to = c("sex", ".value"),
    names_sep = "s "
  ) |>
  rename(age = `gestational age`) |>
  mutate(
    id = row_number(),
    sex = sex |>
      case_match(
        "boy" ~ "male",
        "girl" ~ "female"
      ) |>
      factor(levels = c("female", "male")),
    male = sex == "male",
    female = sex == "female"
  )

bw

#> # A tibble: 24 x 6
#>   sex      age weight    id male  female
#>   <fct> <dbl> <dbl> <int> <lgl> <lgl>
#> 1 male    40   2968     1 TRUE  FALSE
#> 2 female  40   3317     2 FALSE TRUE
#> 3 male    38   2795     3 TRUE  FALSE
#> 4 female  36   2729     4 FALSE TRUE
#> 5 male    40   3163     5 TRUE  FALSE
#> 6 female  40   2935     6 FALSE TRUE
#> 7 male    35   2925     7 TRUE  FALSE
#> 8 female  38   2754     8 FALSE TRUE
#> 9 male    36   2625     9 TRUE  FALSE
#> 10 female 42   3210    10 FALSE TRUE
#> # i 14 more rows
```

2.2.4 Data notation

Let's define some notation to represent this data:

- Y : birthweight (measured in grams)
- S : chromosomal sex: "male" (XY) or "female" (XX)
- M : indicator variable for $S = \text{"male"}$ ²
- $M = 0$ if $S = \text{"female"}$
- $M = 1$ if $S = \text{"male"}$
- F : indicator variable for $S = \text{"female"}$ ³
- $F = 1$ if $S = \text{"female"}$
- $F = 0$ if $S = \text{"male"}$
- A : estimated gestational age at birth (measured in weeks).

Female is the **reference level** for the categorical variable S (chromosomal sex) and corresponding indicator variable M . The choice of a reference level is arbitrary and does not limit what we can do with the resulting model; it only makes it more computationally convenient to make inferences about comparisons involving that reference group.

M and F are called **dummy variables**; together, they are a numeric representation of the categorical variable S . Dummy variables with values 0 and 1 are also called **indicator variables**. There are other ways to construct dummy variables, such as using the values -1 and 1 (see Dobson and Barnett (2018) §2.4 for details).

2.3 Parallel lines regression

(c.f. Dunn and Smyth (2018) §2.10.3⁴)

We don't have enough data to model the distribution of birth weight separately for each combination of gestational age and sex, so let's instead consider a (relatively) simple model for how that distribution varies with gestational age and sex:

$$Y|M, A \sim_{\text{ciid}} N(\mu(M, A), \sigma^2)$$
$$\mu(m, a) = \beta_0 + \beta_M m + \beta_A a \quad (1)$$

Table 3 shows the parameter estimates from R. Figure 2 shows the estimated model, superimposed on the data.

```
bw_lm1 <- lm(
  formula = weight ~ sex + age,
  data = bw
)

library(parameters)
bw_lm1 |>
  parameters::parameters() |>
  parameters::print_md(
    include_reference = include_reference_lines,
    select = "{estimate}"
  )
```

² M is implicitly a deterministic function of S

³ F is implicitly a deterministic function of S

⁴https://link.springer.com/chapter/10.1007/978-1-4419-0118-7_2#Sec31

Table 3: Regression parameter estimates for Model 1 of birthweight data

Parameter	Coefficient
(Intercept)	-1773.32
sex (male)	163.04
age	120.89

```
bw <-
  bw |>
  mutate(`E[Y|X=x]` = fitted(bw_lm1)) |>
  arrange(sex, age)

plot2 <-
  plot1 %>% bw +
  geom_line(aes(y = `E[Y|X=x]`))

print(plot2)
```

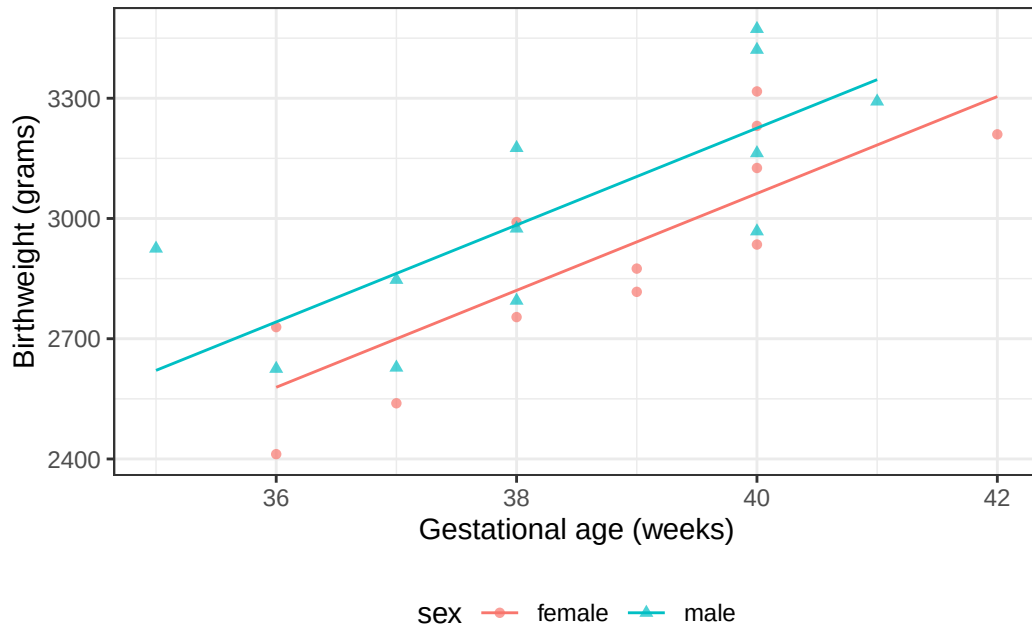


Figure 2: Graph of Model 1 for birthweight data

2.3.1 Model assumptions and predictions

To learn what this model is assuming, let's plug in a few values.

Exercise 2.1. What's the mean birthweight for a female born at 36 weeks?

Table 4: Estimated coefficients for model 1

Parameter	Coefficient
(Intercept)	-1773.32
sex (male)	163.04

Parameter	Coefficient
age	120.89

Solution.

Table 5: Estimated coefficients for model 1

Parameter	Coefficient
(Intercept)	-1773.32
sex (male)	163.04
age	120.89

```
pred_female <- coef(bw_lm1)["(Intercept)"] + coef(bw_lm1)["age"] * 36
## or using built-in prediction:
pred_female_alt <- predict(bw_lm1, newdata = tibble(sex = "female", age = 36))
```

$$\begin{aligned}
 E[Y|M = 0, A = 36] &= \beta_0 + (\beta_M \cdot 0) + (\beta_A \cdot 36) \\
 &= -1773.321839 + (163.039303 \cdot 0) + (120.894327 \cdot 36) \\
 &= 2578.873934
 \end{aligned}$$

Exercise 2.2. What's the mean birthweight for a male born at 36 weeks?

Table 6: Estimated coefficients for model 1

Parameter	Coefficient
(Intercept)	-1773.32
sex (male)	163.04
age	120.89

Solution.

Table 7: Estimated coefficients for model 1

Parameter	Coefficient
(Intercept)	-1773.32
sex (male)	163.04
age	120.89

```
pred_male <-
  coef(bw_lm1)["(Intercept)"] +
  coef(bw_lm1)["sexmale"] +
  coef(bw_lm1)["age"] * 36
```

$$\begin{aligned}
 E[Y|M = 1, A = 36] &= \beta_0 + \beta_M \cdot 1 + \beta_A \cdot 36 \\
 &= 2741.913237
 \end{aligned}$$

Exercise 2.3. What's the difference in mean birthweights between males born at 36 weeks and females born at 36 weeks?

```
coef(bw_lm1)
#> (Intercept)      sexmale      age
#>   -1773.322    163.039    120.894
```

Solution.

$$\begin{aligned} E[Y|M=1, A=36] - E[Y|M=0, A=36] \\ &= 2741.913237 - 2578.873934 \\ &= 163.039303 \end{aligned}$$

Shortcut:

$$\begin{aligned} E[Y|M=1, A=36] - E[Y|M=0, A=36] \\ &= (\beta_0 + \beta_M \cdot 1 + \beta_A \cdot 36) - (\beta_0 + \beta_M \cdot 0 + \beta_A \cdot 36) \\ &= \beta_M \\ &= 163.039303 \end{aligned}$$

Age cancels out in this difference. In other words, according to this model, the difference between females and males with the same gestational age is the same for every age.

This characteristic is an assumption of the model specified by Equation 1. It's hardwired into the parametric model structure, even before we estimated values for those parameters.

2.3.2 Coefficient Interpretation

Recall Model 1:

$$E[Y|M=m, A=a] = \mu(m, a) = \beta_0 + \beta_M m + \beta_A a$$

Slope (of the mean with respect to age) for males:

$$\begin{aligned} \frac{d}{da} \mu(1, a) &= \frac{d}{da} (\beta_0 + \beta_M 1 + \beta_A a) \\ &= \left(\frac{d}{da} \beta_0 + \frac{d}{da} \beta_M 1 + \frac{d}{da} \beta_A a \right) \\ &= (0 + 0 + \beta_A) \\ &= \beta_A \end{aligned}$$

Slope for females:

$$\begin{aligned} \frac{d}{da} \mu(0, a) &= \frac{d}{da} (\beta_0 + \beta_M 0 + \beta_A a) \\ &= \left(\frac{d}{da} \beta_0 + \frac{d}{da} \beta_M 0 + \frac{d}{da} \beta_A a \right) \\ &= (0 + 0 + \beta_A) \\ &= \beta_A \end{aligned}$$

Exercise 2.4. What is the interpretation of β_A in Model 1?

Solution.

$$\begin{aligned}
 \frac{d}{da}\mu(m, a) &= \frac{d}{da}(\beta_0 + \beta_M m + \beta_A a) \\
 &= \left(\frac{d}{da}\beta_0 + \frac{d}{da}\beta_M m + \frac{d}{da}\beta_A a \right) \\
 &= (0 + 0 + \beta_A) \\
 &= \beta_A
 \end{aligned}$$

Conclusion:

$$\beta_A = \frac{d}{da}\mu(m, a)$$

β_A is the slope of mean birthweight with respect to gestational age, adjusting for sex.

Or we can plug in the definition of slope:

$$\beta_A = E[Y|M = m, A = a + 1] - E[Y|M = m, A = a]$$

Exchangeability and consistency have not been assessed; so we are not discussing potential outcomes (causality), only observed outcomes.

Exercise 2.5. What is the interpretation of β_M in Model 1?

Solution.

More precisely written:

$$E[Y|M = m, A = a] = \mu(m, a) = \begin{cases} \beta_0 + \beta_M m + \beta_A a, & \text{for } m \in \{0, 1\} \\ \text{undefined}, & \text{for } m \notin \{0, 1\} \end{cases}$$

The model is undefined for $m \notin \{0, 1\}$, so the derivative with respect to m doesn't exist.

$$\begin{aligned}
 E[Y|M = 1, A = a] &= \beta_0 + \beta_M 1 + \beta_A a \\
 &= \beta_0 + \beta_M + \beta_A a \\
 E[Y|M = 0, A = a] &= \beta_0 + \beta_M 0 + \beta_A a \\
 &= \beta_0 + \beta_A a
 \end{aligned}$$

So:

$$\begin{aligned}
 E[Y|M = 1, A = a] - E[Y|M = 0, A = a] &= (\beta_0 + \beta_M + \beta_A a) - (\beta_0 + \beta_A a) \\
 &= \beta_M
 \end{aligned}$$

Therefore:

$$\begin{aligned}
 \beta_M &= E[Y|M = 1, A = a] - E[Y|M = 0, A = a] \\
 &= \mu(1, a) - \mu(0, a)
 \end{aligned}$$

In words: β_M is the difference in mean birthweight between males and females adjusting for age.

Exercise 2.6. $\beta_0 = ?$

Solution.

$$\begin{aligned}
 E[Y|M = 0, A = 0] &= \mu(0, 0) \\
 &= \beta_0 + \beta_M 0 + \beta_A 0 \\
 &= \beta_0 \\
 \beta_0 &= E[Y|M = 0, A = 0] = \mu(0, 0)
 \end{aligned}$$

β_0 is the mean birthweight for a female with gestational age 0 weeks.

2.4 Interactions

What if we don't like that parallel lines assumption?

Then we need to allow an "interaction" between age A and sex S :

$$E[Y|S = s, A = a] = \beta_0 + \beta_A a + \beta_M m + \beta_{AM}(a \cdot m) \quad (2)$$

Now, the slope of mean birthweight $E[Y|A, S]$ with respect to gestational age A depends on the value of sex S .

```
bw_lm2 <- lm(weight ~ sex + age + sex:age, data = bw)
bw_lm2 |>
  parameters() |>
  parameters::print_md(
    include_reference = include_reference_lines,
    select = "{estimate}"
  )
```

Table 8: Birthweight model with interaction term

Parameter	Coefficient
(Intercept)	-2141.67
sex (male)	872.99
age	130.40
sex (male) $\times$ age	-18.42

```
bw <-
  bw |>
  mutate(
    predlm2 = predict(bw_lm2)
  ) |>
  arrange(sex, age)

plot1_interact <-
  plot1 %>% bw +
  geom_line(aes(y = predlm2))

print(plot1_interact)
```

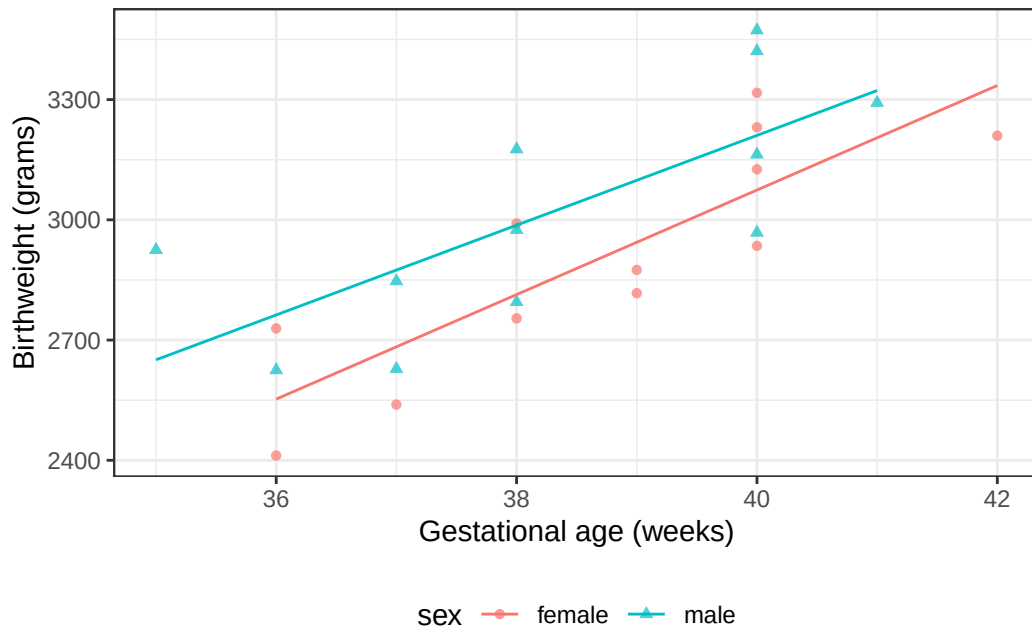


Figure 3: Birthweight model with interaction term

Now we can see that the lines aren't parallel.

Here's another way we could rewrite this model (by collecting terms involving S):

$$E[Y|M, A] = \beta_0 + \beta_M M + (\beta_A + \beta_{AM} M)A$$

If you want to understand a coefficient in a model with interactions, collect terms for the corresponding variable, and you will see which other covariates interact with the variable whose coefficient you are interested in. In this case, the association between A (age) varies between males and females (that is, by sex S).⁵ So the slope of Y with respect to A depends on the value of M . According to this model, there is no such thing as “the slope of birthweight with respect to age”. There are two slopes, one for each sex. We can only talk about “the slope of birthweight with respect to age among males” and “the slope of birthweight with respect to age among females”. Then: each non-interaction slope coefficient is the difference in means per unit difference in its corresponding variable, when all interacting variables are set to 0.

To learn what this model is assuming, let's plug in a few values.

Exercise 2.7. According to this model, what's the mean birthweight for a female born at 36 weeks?

Parameter	Coefficient
(Intercept)	-2141.67
sex (male)	872.99
age	130.40
sex (male) $\times$ age	-18.42

⁵some call this kind of variation “interaction” or “effect modification”, but “act”, “effect”, “modify”, and “by” all suggest causality, which we are not prepared to assess here; let's try to avoid using causal terms, unless we are constructing a causal model.

Solution.

Parameter	Coefficient
(Intercept)	-2141.67
sex (male)	872.99
age	130.40
sex (male) $\times$ age	-18.42

```
pred_female <- coef(bw_lm2)["(Intercept)"] + coef(bw_lm2)["age"] * 36
```

$$E[Y|M = 0, A = 36] = \beta_0 + \beta_M \cdot 0 + \beta_A \cdot 36 + \beta_{AM} \cdot (0 \cdot 36) = 2552.733333$$

Exercise 2.8. What's the mean birthweight for a male born at 36 weeks?

Parameter	Coefficient
(Intercept)	-2141.67
sex (male)	872.99
age	130.40
sex (male) $\times$ age	-18.42

Solution.

Parameter	Coefficient
(Intercept)	-2141.67
sex (male)	872.99
age	130.40
sex (male) $\times$ age	-18.42

```
pred_male <-  
  coef(bw_lm2)["(Intercept)"] +  
  coef(bw_lm2)["sexmale"] +  
  coef(bw_lm2)["age"] * 36 +  
  coef(bw_lm2)["sexmale:age"] * 36
```

$$\begin{aligned} E[Y|M = 1, A = 36] &= \beta_0 + \beta_M \cdot 1 + \beta_A \cdot 36 + \beta_{AM} \cdot 1 \cdot 36 \\ &= 2762.706897 \end{aligned}$$

Exercise 2.9. What's the difference in mean birthweights between males born at 36 weeks and females born at 36 weeks?

Solution.

$$\begin{aligned} &E[Y|M = 1, A = 36] - E[Y|M = 0, A = 36] \\ &= (\beta_0 + \beta_M \cdot 1 + \beta_A \cdot 36 + \beta_{AM} \cdot 1 \cdot 36) \\ &\quad - (\beta_0 + \beta_M \cdot 0 + \beta_A \cdot 36 + \beta_{AM} \cdot 0 \cdot 36) \\ &= \beta_M + \beta_{AM} \cdot 36 \\ &= 209.973563 \end{aligned}$$

Note that age now does show up in the difference: in other words, according to this model, the difference in mean birthweights between females and males with the same gestational age can vary by gestational age.

That's how the lines in the graph ended up non-parallel.

2.4.1 Coefficient Interpretation

Exercise 2.10. What is the interpretation of β_M in Model 2?

Solution.

Mean birthweight among males with gestational age 0 weeks:

$$\begin{aligned}\mu(1, 0) &= E[Y|M = 1, A = 0] \\ &= \beta_0 + \beta_M \cdot 1 + \beta_A \cdot 0 + \beta_{AM} \cdot 1 \cdot 0 \\ &= \beta_0 + \beta_M\end{aligned}$$

Mean birthweight among females with gestational age 0 weeks:

$$\begin{aligned}\mu(0, 0) &= E[Y|M = 0, A = 0] \\ &= \beta_0 + \beta_M \cdot 0 + \beta_A \cdot 0 + \beta_{AM} \cdot 0 \cdot 0 \\ &= \beta_0\end{aligned}$$

$$\begin{aligned}\beta_M &= \mu(1, 0) - \mu(0, 0) \\ &= E[Y|M = 1, A = 0] - E[Y|M = 0, A = 0]\end{aligned}$$

β_M is the difference in mean birthweight between males with gestational age 0 weeks and females with gestational age 0 weeks.

Exercise 2.11. What is the interpretation of β_{AM} in Model 2?

Solution.

Slope among males:

$$\begin{aligned}\frac{\partial}{\partial a}\mu(1, a) &= \frac{\partial}{\partial a}(\beta_0 + \beta_M \cdot 1 + \beta_A \cdot a + \beta_{AM} \cdot 1 \cdot a) \\ &= \frac{\partial}{\partial a}(\beta_0 + \beta_M + \beta_A \cdot a + \beta_{AM} \cdot a) \\ &= \beta_A + \beta_{AM}\end{aligned}$$

or

$$\begin{aligned}E[Y|1, a+1] - E[Y|1, a] &= \beta_0 + \beta_M \cdot 1 + \beta_A(a+1) + \beta_{AM} \cdot 1(a+1) \\ &\quad - (\beta_0 + \beta_M \cdot 1 + \beta_A(a) + \beta_{AM} \cdot 1(a)) \\ &= \beta_A + \beta_{AM}\end{aligned}$$

Slope among females:

$$\begin{aligned}\frac{\partial}{\partial a}\mu(0, a) &= \frac{\partial}{\partial a}(\beta_0 + \beta_M \cdot 0 + \beta_A \cdot a + \beta_{AM} \cdot 0 \cdot a) \\ &= \frac{\partial}{\partial a}(\beta_0 + \beta_A \cdot a) \\ &= \beta_A\end{aligned}$$

or

$$\begin{aligned}
E[Y|0, a+1] - E[Y|0, a] &= \beta_0 + \beta_M 0 + \beta_A(a+1) + \beta_{AM} 0(a+1) \\
&\quad - (\beta_0 + \beta_M 0 + \beta_A(a) + \beta_{AM} 0(a)) \\
&= \beta_0 + \beta_A(a+1) - (\beta_0 + \beta_A(a)) \\
&= \beta_A
\end{aligned}$$

Difference in slopes:

$$\begin{aligned}
\frac{\partial}{\partial a} \mu(1, a) - \frac{\partial}{\partial a} \mu(0, a) &= \beta_A + \beta_{AM} - \beta_A \\
&= \beta_{AM}
\end{aligned}$$

or

$$\begin{aligned}
(E[Y|1, a+1] - E[Y|1, a]) - (E[Y|0, a+1] - E[Y|0, a]) &= \beta_A + \beta_{AM} - \beta_A \\
&= \beta_{AM}
\end{aligned}$$

Therefore

$$\begin{aligned}
\beta_{AM} &= \frac{\partial}{\partial a} \mu(1, a) - \frac{\partial}{\partial a} \mu(0, a) \\
&= (E[Y|M=1, A=a+1] - E[Y|M=1, A=a]) \\
&\quad - (E[Y|M=0, A=a+1] - E[Y|M=0, A=a])
\end{aligned}$$

β_{AM} is the difference in slope of mean birthweight with respect to gestational age between males and females.

2.4.2 Compare coefficient interpretations

Table 9: Coefficient interpretations, by model structure

$\mu(m, a)$	$\beta_0 + \beta_M m + \beta_A a$	$\beta_0 + \beta_M m + \beta_A a + \beta_{AM} ma$
β_0	$\mu(0, 0)$	$\mu(0, 0)$
β_A	$\frac{\partial}{\partial a} \mu(m, a)$	$\frac{\partial}{\partial a} \mu(0, a)$
β_M	$\mu(1, a) - \mu(0, a)$	$\mu(1, 0) - \mu(0, 0)$
β_{AM}		$\frac{\partial}{\partial a} \mu(1, a) - \frac{\partial}{\partial a} \mu(0, a)$

In the model with an interaction term multiplying $A \times M$, the interpretation of β_A involves the reference level of M , and interpretation of β_M involves the reference level of A (Table 9).

2.5 Stratified regression

We could re-write the interaction model as a stratified model, with a slope and intercept for each sex:

$$E[Y|A = a, S = s] = \beta_M m + \beta_{AM}(a \cdot m) + \beta_F f + \beta_{AF}(a \cdot f) \quad (3)$$

Compare this stratified model (Equation 3) with our interaction model, Equation 2:

$$E[Y|A = a, S = s] = \beta_0 + \beta_A a + \beta_M m + \beta_{AM}(a \cdot m)$$

In the stratified model, the intercept term β_0 has been relabeled as β_F .

```

bw_lm2 <- lm(weight ~ sex + age + sex:age, data = bw)
bw_lm2 |>
  parameters() |>
  print_md(

```



```

include_reference = include_reference_lines,
select = "{estimate}"
)

```

Table 10: Birthweight model with interaction term

Parameter	Coefficient
(Intercept)	-2141.67
sex (male)	872.99
age	130.40
sex (male) $\times$ age	-18.42

```

bw_lm_strat <-
  bw |>
  lm(
    formula = weight ~ sex + sex:age - 1,
    data = _
  )

bw_lm_strat |>
  parameters() |>
  print_md(
    select = "{estimate}"
  )

```

Table 11: Birthweight model - stratified betas

Parameter	Coefficient
sex (female)	-2141.67
sex (male)	-1268.67
sex (female) $\times$ age	130.40
sex (male) $\times$ age	111.98

2.6 Curved-line regression

If we transform some of our covariates (X s) and plot the resulting model on the original covariate scale, we end up with curved regression lines:

```

bw_lm3 <- lm(weight ~ sex:log(age) - 1, data = bw)

ggbw <-
  bw |>
  ggplot(
    aes(x = age, y = weight)
  ) +
  geom_point() +
  xlab("Gestational Age (weeks)") +
  ylab("Birth Weight (g)")

ggbw2 <- ggbw +
  stat_smooth(
    method = "lm",
    formula = y ~ log(x),
    geom = "smooth"
  ) +
  xlab("Gestational Age (weeks)") +

```

```
ylab("Birth Weight (g)")

ggbw2 |> print()
```

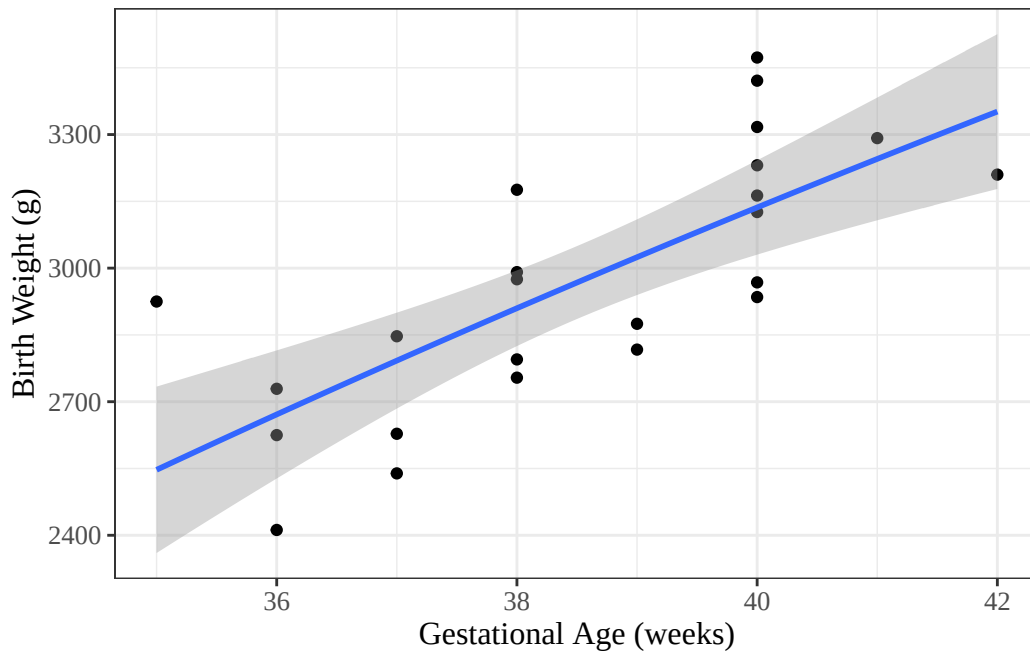


Figure 4: `birthweight` model with `age` entering on log scale

Below is an example with a slightly more obvious curve.

```
library(palmerpenguins)

ggpenguins <-
  palmerpenguins::penguins |>
  dplyr::filter(species == "Adelie") |>
  ggplot(
    aes(x = bill_length_mm, y = body_mass_g)
  ) +
  geom_point() +
  xlab("Bill length (mm)") +
  ylab("Body mass (g)")

ggpenguins2 <- ggpenguins +
  stat_smooth(
    method = "lm",
    formula = y ~ log(x),
    geom = "smooth"
  ) +
  xlab("Bill length (mm)") +
  ylab("Body mass (g)")

ggpenguins2 |> print()
```

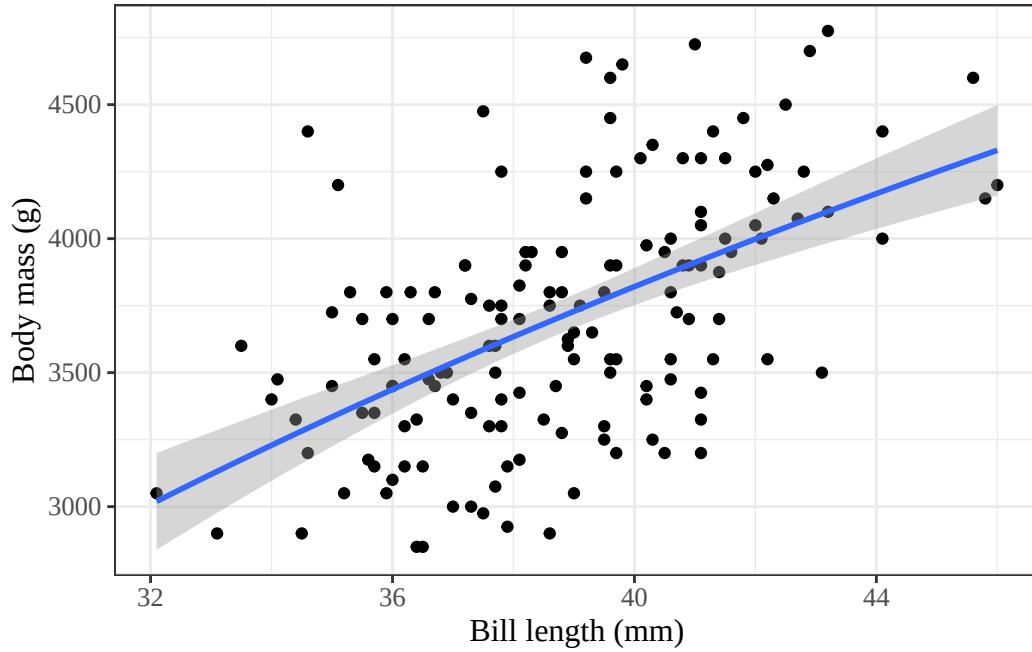


Figure 5: palmerpenguins model with bill_length entering on log scale

2.7 Rescaling

Centering covariates (subtracting a constant from them) can make coefficient interpretations more meaningful, particularly for the intercept and main effect terms in models with interactions.

2.7.1 Rescale age

For example, the intercept β_0 in our interaction model represents the mean birthweight for females at age = 0 weeks, which is not a biologically plausible scenario. If we center age at 36 weeks instead, β_0 will represent the mean birthweight for females at 36 weeks, which is more interpretable.

Exercise 2.12. Let $A^* = A - 32$ weeks.

Consider a new version of Model 2, with A^* in place of A :

$$E[Y|M = m, A^* = a^*] = \gamma_0 + \gamma_M m + \gamma_{A^*} a^* + \gamma_{A^*M} (m \cdot a^*) \quad (4)$$

Let the coefficients of this model be γ s instead of β s.

What are the interpretations of the γ s? How do they relate to the β s in Model 2? Which have the same interpretation? Which are different, and how do they differ? What is the pattern?

Solution. **Interpretation of γ_0 :**

From Model 4, γ_0 is the mean birthweight among females ($M = 0$) with $A^* = 0$ (i.e., $A = 32$ weeks):

$$\gamma_0 = E[Y|M = 0, A^* = 0] = E[Y|M = 0, A = 32]$$

Substituting into Model 2:

$$\begin{aligned} \gamma_0 &= \beta_0 + \beta_M \cdot 0 + \beta_A \cdot 32 + \beta_{AM} \cdot 0 \cdot 32 \\ &= \beta_0 + 32\beta_A \end{aligned}$$

This differs from $\beta_0 = E[Y|M = 0, A = 0]$, which is the mean birthweight among females at $A = 0$ weeks.

Interpretation of γ_M :

From Model 4, γ_M is the sex difference in mean birthweight at $A^* = 0$ (i.e., $A = 32$ weeks):

$$\begin{aligned}\gamma_M &= E[Y|M = 1, A^* = 0] - E[Y|M = 0, A^* = 0] \\ &= E[Y|M = 1, A = 32] - E[Y|M = 0, A = 32]\end{aligned}$$

Substituting into Model 2:

$$\begin{aligned}\gamma_M &= (\beta_0 + \beta_M + \beta_A \cdot 32 + \beta_{AM} \cdot 32) - (\beta_0 + \beta_A \cdot 32) \\ &= \beta_M + 32\beta_{AM}\end{aligned}$$

This differs from β_M , which is the sex difference at $A = 0$ weeks.

Interpretation of γ_{A^*} :

From Model 4, γ_{A^*} is the slope of mean birthweight with respect to A^* among females ($M = 0$):

$$\begin{aligned}\gamma_{A^*} &= \frac{d}{da^*} E[Y|M = 0, A^* = a^*] \\ &= \frac{d}{da^*} E[Y|M = 0, A = a^* + 32] \\ &= \frac{d}{da} E[Y|M = 0, A = a]\end{aligned}$$

Substituting into Model 2:

$$\begin{aligned}\gamma_{A^*} &= \frac{d}{da} (\beta_0 + \beta_A \cdot a) \\ &= \beta_A\end{aligned}$$

Since shifting A by a constant does not change the slope, $\gamma_{A^*} = \beta_A$: these two coefficients have the same value and interpretation.

Interpretation of γ_{A^*M} :

From Model 4, γ_{A^*M} is the difference in slope with respect to A^* between males and females:

$$\begin{aligned}\gamma_{A^*M} &= \frac{d}{da^*} E[Y|M = 1, A^* = a^*] - \frac{d}{da^*} E[Y|M = 0, A^* = a^*] \\ &= \frac{d}{da} E[Y|M = 1, A = a] - \frac{d}{da} E[Y|M = 0, A = a]\end{aligned}$$

Substituting into Model 2:

$$\begin{aligned}\gamma_{A^*M} &= (\beta_A + \beta_{AM}) - \beta_A \\ &= \beta_{AM}\end{aligned}$$

Since shifting A by a constant does not change slopes, $\gamma_{A^*M} = \beta_{AM}$: these two coefficients have the same value and interpretation.

The pattern:

Slope coefficients (γ_{A^*} and γ_{A^*M}) are unchanged by rescaling: they have the same values and interpretations as the corresponding β s.

Coefficients change only for variables that have interactions with the rescaled variable A . This includes the intercept (which can be viewed as the main effect of a variable that interacts with A via

β_A), and the main effect of M (which interacts with A via β_{AM}). Shifting A by 32 weeks changes the reference point from $A = 0$ to $A = 32$, so these coefficients now represent quantities evaluated at $A = 32$ weeks rather than at $A = 0$ weeks.

Exercise 2.13. Using R, fit the rescaled interaction model with $A^* = A - 36$ weeks in place of A in Model 2. Compare the coefficient estimates with those from the original model. Which coefficients change, and which remain the same?

Solution.

```
bw <-
  bw |>
  mutate(
    `age - mean` = age - mean(age),
    `age - 36wks` = age - 36
  )

lm1_c <- lm(weight ~ sex + `age - 36wks`, data = bw)

lm2_c <- lm(weight ~ sex + `age - 36wks` + sex:`age - 36wks`, data = bw)

parameters(lm2_c, ci_method = "wald") |> print_md()
```

Parameter	Coefficient	SE	95% CI	t(20)	p
(Intercept)	2552.73	97.59	(2349.16, 2756.30)	26.16	< .001
sex (male)	209.97	129.75	(-60.68, 480.63)	1.62	0.121
age - 36wks	130.40	30.00	(67.82, 192.98)	4.35	< .001
sex (male) $\times$ age - 36wks	-18.42	41.76	(-105.52, 68.68)	-0.44	0.664

Compare with what we got without rescaling:

```
parameters(bw_lm2, ci_method = "wald") |> print_md()
```

Parameter	Coefficient	SE	95% CI	t(20)	p
(Intercept)	-2141.67	1163.60	(-4568.90, 285.56)	-1.84	0.081
sex (male)	872.99	1611.33	(-2488.18, 4234.17)	0.54	0.594
age	130.40	30.00	(67.82, 192.98)	4.35	< .001
sex (male) $\times$ age	-18.42	41.76	(-105.52, 68.68)	-0.44	0.664

Notice that the slope coefficients (`age - 36wks` and `sexmale:age - 36wks`) remain the same, but the intercept and `sexmale` coefficient have changed to reflect means at age = 36 weeks rather than age = 0 weeks.

3 Estimating Linear Models via Maximum Likelihood

In EPI 203 and our review of MLEs⁶, we learned how to fit outcome-only models of the form $p(X = x|\theta)$ to iid data $\tilde{x} = (x_1, \dots, x_n)$ using maximum likelihood estimation.

Now, we apply the same procedure to linear regression models:

⁶[intro-MLEs.qmd#sec-intro-MLEs](#)

3.0.1 Likelihood

$$\begin{aligned}
\mathcal{L}_i &\stackrel{\text{def}}{=} p(Y_i = y_i | \tilde{X}_i = \tilde{x}_i) \\
&= (2\pi\sigma^2)^{-1/2} \exp\left\{-\frac{1}{2\sigma^2}\varepsilon_i^2\right\} \\
\varepsilon_i &\stackrel{\text{def}}{=} y_i - \mu_i \\
\mu_i &\stackrel{\text{def}}{=} \mu(x_i) \\
&= x_i \cdot \beta
\end{aligned}$$

$$\begin{aligned}
\mathcal{L} &\stackrel{\text{def}}{=} \mathcal{L}(\tilde{y}|\mathbf{x}, \tilde{\beta}, \sigma^2) \\
&\stackrel{\text{def}}{=} p(\tilde{Y} = \tilde{y} | \mathbf{X} = \mathbf{x}) \\
&= \prod_{i=1}^n \mathcal{L}_i
\end{aligned} \tag{5}$$

3.0.2 Log-likelihood

$$\begin{aligned}
\ell_i &\stackrel{\text{def}}{=} \log\{\mathcal{L}_i\} \\
&= \log\left\{(2\pi\sigma^2)^{-1/2} \exp\left\{-\frac{1}{2\sigma^2}\varepsilon_i^2\right\}\right\} \\
&= -\frac{1}{2}\log\{2\pi\sigma^2\} - \frac{1}{2\sigma^2}\varepsilon_i^2
\end{aligned}$$

$$\begin{aligned}
\ell &\stackrel{\text{def}}{=} \ell(\tilde{y}|\mathbf{x}, \beta, \sigma^2) \\
&\stackrel{\text{def}}{=} \log\{\mathcal{L}(\tilde{y}|\mathbf{x}, \beta, \sigma^2)\} \\
&= \log\left\{\prod_{i=1}^n \mathcal{L}_i\right\} \\
&= \sum_{i=1}^n \log\{\mathcal{L}_i\} \\
&= \sum_{i=1}^n \ell_i \\
&= \sum_{i=1}^n \left(-\frac{1}{2}\log\{2\pi\sigma^2\} - \frac{1}{2\sigma^2}\varepsilon_i^2\right) \\
&= -\frac{n}{2}\log\{2\pi\sigma^2\} - \frac{1}{2\sigma^2} \sum_{i=1}^n \varepsilon_i^2 \\
&= -\frac{n}{2}\log\{2\pi\sigma^2\} - \frac{1}{2\sigma^2} (\tilde{\varepsilon} \cdot \tilde{\varepsilon}) \\
&= -\frac{n}{2}\log\{2\pi\sigma^2\} - \frac{1}{2\sigma^2} ((\tilde{y} - \tilde{\mu}) \cdot (\tilde{y} - \tilde{\mu})) \\
&= -\frac{n}{2}\log\{2\pi\sigma^2\} - \frac{1}{2\sigma^2} ((\tilde{y} - \mathbf{X}\tilde{\beta}) \cdot (\tilde{y} - \mathbf{X}\tilde{\beta})) \\
&= -\frac{n}{2}\log\{2\pi\sigma^2\} - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - (\tilde{x}_i \cdot \tilde{\beta}))^2
\end{aligned} \tag{6}$$

3.0.3 Score function

$$\begin{aligned}
\mu'_i &\stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{\beta}} \mu_i \\
&= \frac{\partial}{\partial \tilde{\beta}} (\tilde{x}_i \cdot \tilde{\beta}) \\
&= \left(\frac{\partial}{\partial \tilde{\beta}} \tilde{\beta} \right) \tilde{x}_i \\
&= \mathbb{I} \tilde{x}_i \\
&= \tilde{x}_i
\end{aligned}$$

$$\begin{aligned}
\varepsilon'_i &\stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{\beta}} \varepsilon_i \\
&= \frac{\partial}{\partial \tilde{\beta}} (y_i - \mu_i) \\
&= \frac{\partial}{\partial \tilde{\beta}} y_i - \frac{\partial}{\partial \tilde{\beta}} \mu_i \\
&= 0 - \tilde{x}_i \\
&= -\tilde{x}_i
\end{aligned}$$

$$\begin{aligned}
\ell'_i &\stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{\beta}} \ell_i \\
&= \frac{\partial}{\partial \tilde{\beta}} \left(-\frac{1}{2} \log\{2\pi\sigma^2\} - \frac{1}{2\sigma^2} \varepsilon_i^2 \right) \\
&= \frac{\partial}{\partial \tilde{\beta}} \left(-\frac{1}{2} \log\{2\pi\sigma^2\} \right) - \frac{\partial}{\partial \tilde{\beta}} \frac{1}{2\sigma^2} \varepsilon_i^2 \\
&= 0 - \frac{1}{2\sigma^2} \frac{\partial}{\partial \tilde{\beta}} \varepsilon_i^2 \\
&= -\frac{1}{2\sigma^2} 2(\varepsilon'_i) \varepsilon_i \\
&= -\frac{1}{\sigma^2} (-\tilde{x}_i \varepsilon_i) \\
&= \frac{1}{\sigma^2} \tilde{x}_i \varepsilon_i
\end{aligned}$$

$$\begin{aligned}
\ell'_{\tilde{\beta}} &\stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{\beta}} \ell_{\tilde{\beta}} \\
&= \frac{\partial}{\partial \tilde{\beta}} \sum_{i=1}^n \ell_i \\
&= \sum_{i=1}^n \frac{\partial}{\partial \tilde{\beta}} \ell_i \\
&= \sum_{i=1}^n \ell'_i \\
&= \sum_{i=1}^n \frac{1}{\sigma^2} \tilde{x}_i \varepsilon_i \\
&= \frac{1}{\sigma^2} \sum_{i=1}^n \tilde{x}_i \varepsilon_i \\
&= \frac{1}{\sigma^2} \mathbf{X}^\top \tilde{\varepsilon}
\end{aligned}$$

3.0.4 Hessian

$$\begin{aligned}
\ell''_i &\stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{\beta}^\top} \frac{\partial}{\partial \tilde{\beta}} \ell_i \\
&= \frac{\partial}{\partial \tilde{\beta}^\top} \ell'_i \\
&= \frac{\partial}{\partial \tilde{\beta}^\top} \left(\frac{1}{\sigma^2} \tilde{x}_i \varepsilon_i \right) \\
&= \frac{1}{\sigma^2} \tilde{x}_i \varepsilon'_i{}^\top \\
&= \frac{1}{\sigma^2} \tilde{x}_i (-\tilde{x}_i^\top) \\
&= -\frac{1}{\sigma^2} \tilde{x}_i \tilde{x}_i^\top
\end{aligned}$$

$$\begin{aligned}
\ell'' &\stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{\beta}^\top} \frac{\partial}{\partial \tilde{\beta}} \ell \\
&= \frac{\partial}{\partial \tilde{\beta}^\top} \ell' \\
&= \frac{\partial}{\partial \tilde{\beta}^\top} \sum_{i=1}^n \ell'_i \\
&= \sum_{i=1}^n \frac{\partial}{\partial \tilde{\beta}^\top} \ell'_i \\
&= \sum_{i=1}^n \ell''_i \\
&= \sum_{i=1}^n -\frac{1}{\sigma^2} \tilde{x}_i \tilde{x}_i^\top \\
&= -\frac{1}{\sigma^2} \sum_{i=1}^n \tilde{x}_i \tilde{x}_i^\top \\
&= -\frac{1}{\sigma^2} \mathbf{X}^\top \mathbf{X}
\end{aligned}$$

That is,

$$\ell'' = -\frac{1}{\sigma^2} \sum_{i=1}^n \tilde{x}_i \tilde{x}_i^\top \quad (7)$$

3.0.5 Alternative approach using matrix derivatives

$$\begin{aligned} \ell'_{\tilde{\beta}}(\tilde{y}|\mathbf{x}, \tilde{\beta}, \sigma^2) &\stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{\beta}} \ell_{\tilde{\beta}}(\tilde{y}|\mathbf{x}, \tilde{\beta}, \sigma^2) \\ &= -\frac{1}{2\sigma^2} \frac{\partial}{\partial \tilde{\beta}} \left(\sum_{i=1}^n (y_i - (\tilde{x}_i \cdot \tilde{\beta}))^2 \right) \end{aligned} \quad (8)$$

Let's switch to matrix-vector notation:

$$\sum_{i=1}^n (y_i - \tilde{x}_i^\top \tilde{\beta})^2 = (\tilde{y} - \mathbf{X}\tilde{\beta}) \cdot (\tilde{y} - \mathbf{X}\tilde{\beta})$$

So

$$\begin{aligned} (\tilde{y} - \mathbf{X}\tilde{\beta})'(\tilde{y} - \mathbf{X}\tilde{\beta}) &= (\tilde{y}' - \tilde{\beta}'\mathbf{X}')(\tilde{y} - \mathbf{X}\tilde{\beta}) \\ &= \tilde{y}'\tilde{y} - \tilde{\beta}'\mathbf{X}'\tilde{y} - \tilde{y}'\mathbf{X}\tilde{\beta} + \tilde{\beta}'\mathbf{X}'\mathbf{X}\tilde{\beta} \\ &= \tilde{y}'\tilde{y} - 2\tilde{y}'\mathbf{X}\tilde{\beta} + \tilde{\beta}'\mathbf{X}'\mathbf{X}\tilde{\beta} \end{aligned}$$

We will use some results from vector calculus⁷:

$$\begin{aligned} \frac{\partial}{\partial \tilde{\beta}} \left(\sum_{i=1}^n (y_i - x'_i \beta)^2 \right) &= \frac{\partial}{\partial \tilde{\beta}} (\tilde{y} - \mathbf{X}\tilde{\beta})'(\tilde{y} - \mathbf{X}\tilde{\beta}) \\ &= \frac{\partial}{\partial \tilde{\beta}} (\tilde{y}'\tilde{y} - 2\tilde{y}'\mathbf{X}\tilde{\beta} + \tilde{\beta}'\mathbf{X}'\mathbf{X}\tilde{\beta}) \\ &= (-2\mathbf{X}'\tilde{y} + 2\mathbf{X}'\mathbf{X}\tilde{\beta}) \\ &= -2\mathbf{X}'(\tilde{y} - \mathbf{X}\tilde{\beta}) \\ &= -2\mathbf{X}'(\tilde{y} - \mathbf{E}[\tilde{y}]) \\ &= -2\mathbf{X}'\varepsilon(\tilde{y}) \end{aligned} \quad (9)$$

So if $\ell'(\beta, \sigma^2) = 0$, then

$$\begin{aligned} 0 &= (-2\mathbf{X}'\tilde{y} + 2\mathbf{X}'\mathbf{X}\beta) \\ 2\mathbf{X}'\tilde{y} &= 2\mathbf{X}'\mathbf{X}\beta \\ \mathbf{X}'\tilde{y} &= \mathbf{X}'\mathbf{X}\beta \\ (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\tilde{y} &= \beta \end{aligned}$$

Hessian

The Hessian (second derivative matrix) is:

$$\ell''_{\beta, \beta'}(\beta, \sigma^2; \tilde{y}, \mathbf{X}) = -\frac{1}{2\sigma^2} \mathbf{X}'\mathbf{X}$$

$\ell''_{\beta, \beta'}(\beta, \sigma^2; \mathbf{X}, \tilde{y})$ is negative definite at $\beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\tilde{y}$, so $\hat{\beta}_{ML} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\tilde{y}$ is the MLE for β .

⁷[math-prereqs.qmd#sec-vector-calculus](#)

Table 12: Covariance matrix of $\hat{\beta}$ for **birthweight** model 2 (with interaction term)

```
bw_lm2 |> vcov()
#>      (Intercept)  sexmale      age  sexmale:age
#> (Intercept)    1353968 -1353968 -34870.966   34870.966
#> sexmale        -1353968  2596387  34870.966  -67210.974
#> age            -34871    34871    899.896   -899.896
#> sexmale:age     34871    -67211   -899.896   1743.548
```

Similarly (not shown):

$$\hat{\sigma}_{ML}^2 = \frac{1}{n}(Y - X\hat{\beta})'(Y - X\hat{\beta})$$

And

$$\begin{aligned} \mathcal{J}_{\beta} &= E[-\ell''_{\beta,\beta'}(Y|X, \beta, \sigma^2)] \\ &= \frac{1}{\sigma^2} \mathbf{X}'\mathbf{X} \end{aligned}$$

So:

$$\text{Var}(\hat{\beta}) \approx (\mathcal{J}_{\beta})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

and

$$\hat{\beta} \sim N(\beta, \mathcal{J}_{\beta}^{-1})$$

These are all results you have hopefully seen before.

In the Gaussian linear regression case, we also have exact results:

$$\frac{\hat{\beta}_j}{\widehat{\text{se}}(\hat{\beta}_j)} \sim t_{n-p}$$

Example 3.1 (MLEs for birthweight data). In model 2 above, $\hat{\mathcal{J}}(\beta)$ is:

If we take the square roots of the diagonals, we get the standard errors listed in the model output:

```
bw_lm2 |>
  vcov() |>
  diag() |>
  sqrt()
#> (Intercept)      sexmale      age  sexmale:age
#>  1163.6015   1611.3309    29.9983    41.7558

bw_lm2 |>
  parameters() |>
  print_md()
```

Table 13: Estimated model for `birthweight` data with interaction term

Parameter	Coefficient	SE	95% CI	t(20)	p
(Intercept)	-2141.67	1163.60	(-4568.90, 285.56)	-1.84	0.081
sex (male)	872.99	1611.33	(-2488.18, 4234.17)	0.54	0.594
age	130.40	30.00	(67.82, 192.98)	4.35	< .001
sex (male) $\times$ age	-18.42	41.76	(-105.52, 68.68)	-0.44	0.664

So we can do confidence intervals, hypothesis tests, and p-values exactly as in the one-variable case we looked at previously.

3.0.6 Residual Standard Deviation

$\hat{\sigma}$ represents an *estimate* of the *Residual Standard Deviation* parameter, σ . We can extract $\hat{\sigma}$ from the fitted model, using the `sigma()` function:

```
sigma(bw_lm2)
#> [1] 180.613
```

σ is NOT “Residual standard error”

In the `summary.lm()` output, this estimate is labeled as “Residual standard error”:

```
summary(bw_lm2)
#>
#> Call:
#> lm(formula = weight ~ sex + age + sex:age, data = bw)
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -246.7  -138.1   -39.1   176.6   274.3
#>
#> Coefficients:
#>              Estimate Std. Error t value Pr(>|t|)
#> (Intercept)  -2141.7      1163.6   -1.84  0.08057 .
#> sexmale         873.0      1611.3    0.54  0.59395
#> age            130.4        30.0    4.35  0.00031 ***
#> sexmale:age    -18.4        41.8   -0.44  0.66389
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 181 on 20 degrees of freedom
#> Multiple R-squared:  0.643, Adjusted R-squared:  0.59
#> F-statistic: 12 on 3 and 20 DF, p-value: 0.000101
```

However, this is a misnomer: see note in `?stats::sigma`

4 Inference about Gaussian Linear Regression Models

4.1 Motivating example: `birthweight` data

Research question: is there really an interaction between sex and age?

$$H_0 : \beta_{AM} = 0$$

$$H_A : \beta_{AM} \neq 0$$

$$P(|\hat{\beta}_{AM}| > | -18.417241 | \mid H_0) = ?$$

4.2 Wald tests and CIs

R can give you Wald tests for single coefficients and corresponding CIs:

```
bw_lm2 |>
  parameters() |>
  print_md(
    include_reference = TRUE
  )
```

Parameter	Coefficient	SE	95% CI	t(20)	p
(Intercept)	-2141.67	1163.60	(-4568.90, 285.56)	-1.84	0.081
sex (female)	0.00				
sex (male)	872.99	1611.33	(-2488.18, 4234.17)	0.54	0.594
age	130.40	30.00	(67.82, 192.98)	4.35	< .001
sex (male) × age	-18.42	41.76	(-105.52, 68.68)	-0.44	0.664

To understand what's happening, let's replicate these results by hand for the interaction term.

4.3 P-values

```
bw_lm2 |>
  parameters(keep = "sexmale:age") |>
  print_md(
    include_reference = TRUE
  )
```

Parameter	Coefficient	SE	95% CI	t(20)	p
sex (male) × age	-18.42	41.76	(-105.52, 68.68)	-0.44	0.664

```
beta_hat <- coef(summary(bw_lm2))["sexmale:age", "Estimate"]
se_hat <- coef(summary(bw_lm2))["sexmale:age", "Std. Error"]
dfresid <- bw_lm2$df.residual
t_stat <- abs(beta_hat) / se_hat
pval_t <-
  pt(-t_stat, df = dfresid, lower.tail = TRUE) +
  pt(t_stat, df = dfresid, lower.tail = FALSE)
```

$$\begin{aligned} &P(|\hat{\beta}_{AM}| > |-18.417241| | H_0) \\ &= \Pr\left(\left|\frac{\hat{\beta}_{AM}}{\widehat{SE}(\hat{\beta}_{AM})}\right| > \left|\frac{-18.417241}{41.755817}\right| \middle| H_0\right) \\ &= \Pr(|T_{20}| > 0.44107 | H_0) \\ &= 0.663893 \end{aligned}$$

This matches the result in the table above.

4.4 Confidence intervals

```
bw_lm2 |>
  parameters(keep = "sexmale:age") |>
  print_md(
    include_reference = TRUE
  )
```

Parameter	Coefficient	SE	95% CI	t(20)	p
sex (male) $\times$ age	-18.42	41.76	(-105.52, 68.68)	-0.44	0.664

```

q_t_upper <- qt(
  p = 0.975,
  df = dfresid,
  lower.tail = TRUE
)

q_t_lower <- qt(
  p = 0.025,
  df = dfresid,
  lower.tail = TRUE
)

confint_radius_t <-
  se_hat * q_t_upper

confint_t <- beta_hat + c(-1, 1) * confint_radius_t

print(confint_t)
#> [1] -105.5184  68.6839

```

This also matches.

4.5 Gaussian approximations

Here are the asymptotic (Gaussian approximation) equivalents:

4.6 P-values

```

bw_lm2 |>
  parameters(keep = "sexmale:age") |>
  print_md(
    include_reference = TRUE
  )

```

Parameter	Coefficient	SE	95% CI	t(20)	p
sex (male) $\times$ age	-18.42	41.76	(-105.52, 68.68)	-0.44	0.664

```

pval_z <- pnorm(abs(t_stat), lower = FALSE) * 2

print(pval_z)
#> [1] 0.659162

```

4.7 Confidence intervals

```

bw_lm2 |>
  parameters(keep = "sexmale:age") |>
  print_md(
    include_reference = TRUE
  )

```

Parameter	Coefficient	SE	95% CI	t(20)	p
sex (male) $\times$ age	-18.42	41.76	(-105.52, 68.68)	-0.44	0.664

```

confint_radius_z <- se_hat * qnorm(0.975, lower = TRUE)
confint_z <-
  beta_hat + c(-1, 1) * confint_radius_z
print(confint_z)
#> [1] -100.2571  63.4227

```

4.8 Likelihood ratio statistics

```

logLik(bw_lm2)
#> 'log Lik.' -156.579 (df=5)
logLik(bw_lm1)
#> 'log Lik.' -156.695 (df=4)

log_LR <- (logLik(bw_lm2) - logLik(bw_lm1)) |> as.numeric()
delta_df <- (bw_lm1$df.residual - df.residual(bw_lm2))

x_max <- 1

```

```

d_log_LR <- function(x, df = delta_df) dchisq(x, df = df)

chisq_plot <-
  ggplot() +
  geom_function(fun = d_log_LR) +
  stat_function(
    fun = d_log_LR,
    xlim = c(log_LR, x_max),
    geom = "area",
    fill = "gray"
  ) +
  geom_segment(
    aes(
      x = log_LR,
      xend = log_LR,
      y = 0,
      yend = d_log_LR(log_LR)
    ),
    col = "red"
  ) +
  xlim(0.0001, x_max) +
  ylim(0, 4) +
  ylab("p(X=x)") +
  xlab("log(likelihood ratio) statistic [x]") +
  theme_classic()
chisq_plot |> print()

```

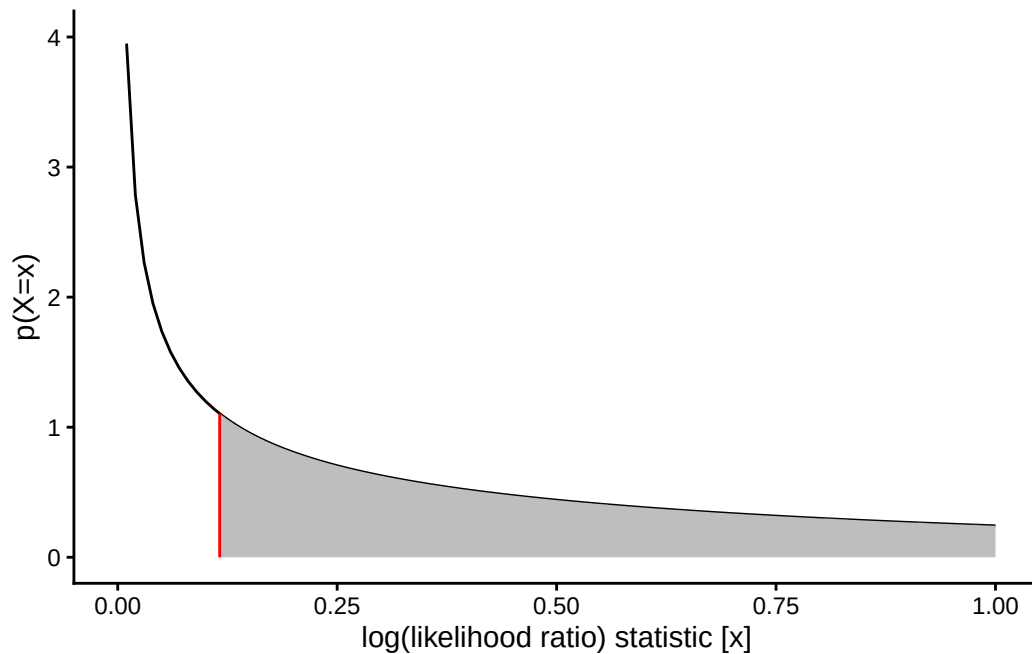


Figure 6: Chi-square distribution

Now we can get the p-value:

```
pchisq(
  q = 2 * log_LR,
  df = delta_df,
  lower = FALSE
) |>
  print()
#> [1] 0.629806
```

In practice you don't have to do this by hand; there are functions to do it for you:

```
# built in
library(lmtest)
lrtest(bw_lm2, bw_lm1)
#> # A tibble: 2 x 5
#>   `#Df` LogLik   Df  Chisq `Pr(>Chisq)`
#>   <dbl> <dbl> <dbl> <dbl>         <dbl>
#> 1     5  -157.   NA  NA           NA
#> 2     4  -157.   -1  0.232         0.630
```

5 Goodness of fit

5.1 AIC and BIC

When we use likelihood ratio tests, we are comparing how well different models fit the data.

Likelihood ratio tests require “nested” models: one must be a special case of the other.

If we have non-nested models, we can instead use the Akaike Information Criterion (AIC) or Bayesian Information Criterion (BIC):

- $AIC = -2 * \ell(\hat{\theta}) + 2 * p$

- $\text{BIC} = -2 * \ell(\hat{\theta}) + p * \log(n)$

where ℓ is the log-likelihood of the data evaluated using the parameter estimates $\hat{\theta}$, p is the number of estimated parameters in the model (including $\hat{\sigma}^2$), and n is the number of observations.

You can calculate these criteria using the `logLik()` function, or use the built-in R functions:

5.1.1 AIC in R

```
-2 * logLik(bw_lm2) |> as.numeric() +
  2 * (length(coef(bw_lm2)) + 1) # sigma counts as a parameter here
#> [1] 323.159

AIC(bw_lm2)
#> [1] 323.159
```

5.1.2 BIC in R

```
-2 * logLik(bw_lm2) |> as.numeric() +
  (length(coef(bw_lm2)) + 1) * log(nobs(bw_lm2))
#> [1] 329.049

BIC(bw_lm2)
#> [1] 329.049
```

Large values of AIC and BIC are worse than small values. There are no hypothesis tests or p-values associated with these criteria.

To compare models using AIC or BIC, calculate the criterion for each model and choose the model with the smallest value. A difference of more than 2 units in AIC is often considered meaningful, though this is just a rough guideline. BIC tends to favor simpler models than AIC, especially when the sample size is large.

5.2 (Residual) Deviance

Let q be the number of distinct covariate combinations in a data set.

```
bw_X_unique <-
  bw |>
  count(sex, age)

n_unique_bw <- nrow(bw_X_unique)
```

For example, in the `birthweight` data, there are $q = 12$ unique patterns (Table 14).

Definition 5.1 (Replicates). If a given covariate pattern has more than one observation in a dataset, those observations are called **replicates**.

Example 5.1 (Replicates in the `birthweight` data). In the `birthweight` dataset, there are 2 replicates of the combination “female, age 36” (Table 14).

Exercise 5.1 (Replicates in the `birthweight` data). Which covariate pattern(s) in the `birthweight` data has the most replicates?

Table 14: Unique covariate combinations in the `birthweight` data, with replicate counts

```

bw_X_unique
#> # A tibble: 12 x 3
#>   sex      age    n
#>   <fct> <dbl> <int>
#> 1 female    36     2
#> 2 female    37     1
#> 3 female    38     2
#> 4 female    39     2
#> 5 female    40     4
#> 6 female    42     1
#> 7 male      35     1
#> 8 male      36     1
#> 9 male      37     2
#> 10 male     38     3
#> 11 male     40     4
#> 12 male     41     1

```

Solution 5.1 (Replicates in the `birthweight` data). Two covariate patterns are tied for most replicates: males at age 40 weeks and females at age 40 weeks. 40 weeks is the usual length for human pregnancy (Polin et al. (2011)), so this result makes sense.

```

bw_X_unique |> dplyr::filter(n == max(n))
#> # A tibble: 2 x 3
#>   sex      age    n
#>   <fct> <dbl> <int>
#> 1 female    40     4
#> 2 male      40     4

```

5.2.1 Saturated models

The most complicated model we could fit would have one parameter (a mean) for each covariate pattern, plus a variance parameter:

```

lm_max <-
  bw |>
  mutate(age = factor(age)) |>
  lm(
    formula = weight ~ sex:age - 1,
    data = _
  )

lm_max |>
  parameters() |>
  print_md()

```

Table 15: Saturated model for the `birthweight` data

Parameter	Coefficient	SE	95% CI	t(12)	p
sex (male) × age35	2925.00	187.92	(2515.55, 3334.45)	15.56	< .001
sex (female) × age36	2570.50	132.88	(2280.98, 2860.02)	19.34	< .001
sex (male) × age36	2625.00	187.92	(2215.55, 3034.45)	13.97	< .001
sex (female) × age37	2539.00	187.92	(2129.55, 2948.45)	13.51	< .001
sex (male) × age37	2737.50	132.88	(2447.98, 3027.02)	20.60	< .001
sex (female) × age38	2872.50	132.88	(2582.98, 3162.02)	21.62	< .001

Table 15: Saturated model for the `birthweight` data

Parameter	Coefficient	SE	95% CI	t(12)	p
sex (male) $\times$ age38	2982.00	108.50	(2745.60, 3218.40)	27.48	< .001
sex (female) $\times$ age39	2846.00	132.88	(2556.48, 3135.52)	21.42	< .001
sex (female) $\times$ age40	3152.25	93.96	(2947.52, 3356.98)	33.55	< .001
sex (male) $\times$ age40	3256.25	93.96	(3051.52, 3460.98)	34.66	< .001
sex (male) $\times$ age41	3292.00	187.92	(2882.55, 3701.45)	17.52	< .001
sex (female) $\times$ age42	3210.00	187.92	(2800.55, 3619.45)	17.08	< .001

We call this model the **full**, **maximal**, or **saturated** model for this dataset.

```
library(rlang) # defines the `.data` pronoun
plot_PIs_and_CIs <- function(model, data) {
  cis <- model |>
    predict(interval = "confidence") |>
    suppressWarnings() |>
    tibble::as_tibble()
  names(cis) <- paste("ci", names(cis), sep = "_")

  preds <- model |>
    predict(interval = "predict") |>
    suppressWarnings() |>
    tibble::as_tibble()
  names(preds) <- paste("pred", names(preds), sep = "_")
  dplyr::bind_cols(bw, cis, preds) |>
    ggplot2::ggplot() +
    ggplot2::aes(
      x = .data$age,
      y = .data$weight,
      col = .data$sex
    ) +
    ggplot2::geom_point() +
    ggplot2::theme(legend.position = "bottom") +
    ggplot2::geom_line(ggplot2::aes(y = .data$ci_fit)) +
    ggplot2::geom_ribbon(
      ggplot2::aes(
        ymin = .data$pred_lwr,
        ymax = .data$pred_upr
      ),
      alpha = 0.2
    ) +
    ggplot2::geom_ribbon(
      ggplot2::aes(
        ymin = .data$ci_lwr,
        ymax = .data$ci_upr
      ),
      alpha = 0.5
    ) +
    ggplot2::facet_wrap(~sex)
}
```

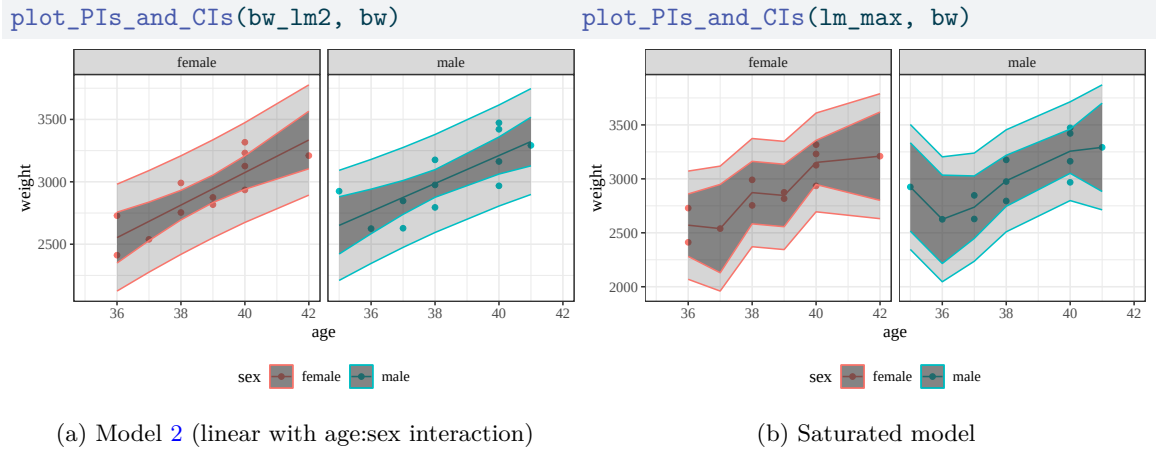


Figure 7: Model 2 and saturated model for `birthweight` data, with confidence and prediction intervals

We can calculate the log-likelihood of this model as usual:

```
logLik(lm_max)
#> 'log Lik.' -151.402 (df=13)
```

We can compare this model to our other models using chi-square tests, as usual:

```
lrtest(lm_max, bw_lm2)
#> # A tibble: 2 x 5
#>   `#Df` LogLik    Df Chisq `Pr(>Chisq)`
#>   <dbl> <dbl> <dbl> <dbl>      <dbl>
#> 1     13  -151.    NA  NA         NA
#> 2      5  -157.    -8  10.4      0.241
```

The likelihood ratio statistic for this test is

$$\lambda = 2 * (\ell_{\text{full}} - \ell) = 10.355374$$

where:

- ℓ_{full} is the log-likelihood of the full model: -151.401601
- ℓ is the log-likelihood of our comparison model (two slopes, two intercepts): -156.579288

This statistic is called the **deviance** or **residual deviance** for our two-slopes and two-intercepts model; it tells us how much the likelihood of that model deviates from the likelihood of the maximal model.

The corresponding p-value tells us whether there we have enough evidence to detect that our two-slopes, two-intercepts model is a worse fit for the data than the maximal model; in other words, it tells us if there's evidence that we missed any important patterns. (Remember, a nonsignificant p-value could mean that we didn't miss anything and a more complicated model is unnecessary, or it could mean we just don't have enough data to tell the difference between these models.)

5.3 Null Deviance

Similarly, the *least* complicated model we could fit would have only one mean parameter, an intercept:

$$E[Y|X = x] = \beta_0$$

We can fit this model in R like so:

```
lm0 <- lm(weight ~ 1, data = bw)

lm0 |>
  parameters() |>
  print_md()
```

Parameter	Coefficient	SE	95% CI	t(23)	p
(Intercept)	2967.67	57.58	(2848.56, 3086.77)	51.54	< .001

```
lm0 |> plot_PIs_and_CIs()
```

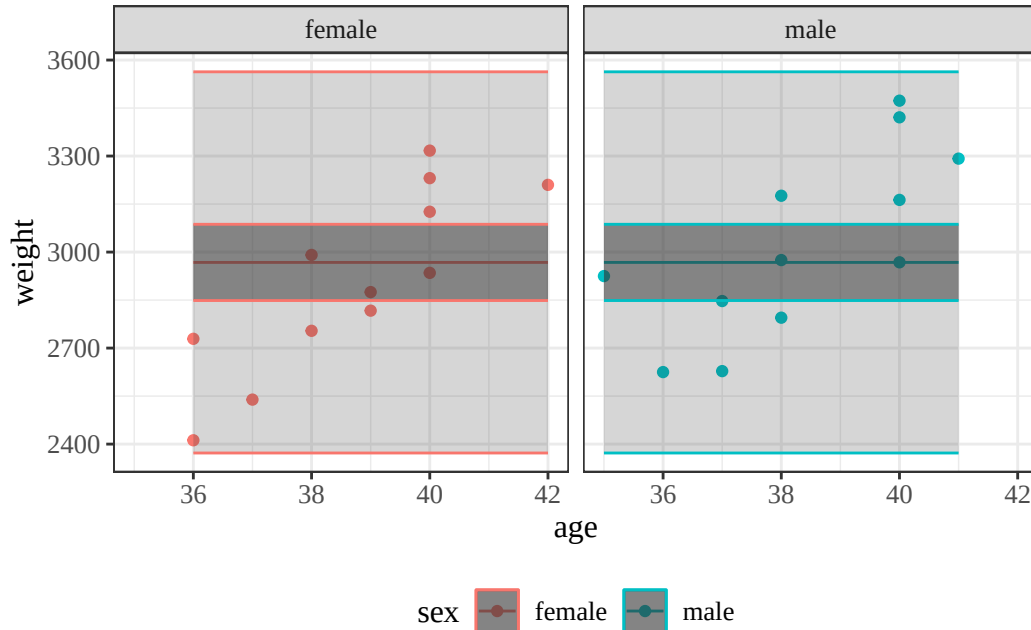


Figure 8: Null model for birthweight data, with 95% confidence and prediction intervals.

This model also has a likelihood:

```
logLik(lm0)
#> 'log Lik.' -168.955 (df=2)
```

And we can compare it to more complicated models using a likelihood ratio test:

```
lrtest(bw_lm2, lm0)
#> # A tibble: 2 x 5
#>   `#Df` LogLik    Df Chisq `Pr(>Chisq)`
#>   <dbl> <dbl> <dbl> <dbl>      <dbl>
#> 1     5  -157.   NA   NA         NA
#> 2     2  -169.   -3  24.8    0.0000174
```

The likelihood ratio statistic for the test comparing the null model to the maximal model is

$$\lambda = 2 * (\ell_{\text{full}} - \ell_0) = 35.106732$$

where:

- ℓ_0 is the log-likelihood of the null model: -168.954967

- ℓ_{full} is the log-likelihood of the maximal model: -151.401601

In R, this test is:

```
lrtest(lm_max, lm0)
#> # A tibble: 2 x 5
#>   `#Df` LogLik    Df Chisq `Pr(>Chisq)`
#>   <dbl> <dbl> <dbl> <dbl>      <dbl>
#> 1     13  -151.    NA    NA         NA
#> 2      2  -169.   -11  35.1    0.000238
```

This log-likelihood ratio statistic is called the **null deviance**. It tells us whether we have enough data to detect a difference between the null and full models.

6 Prediction

6.1 Prediction for linear models

Definition 6.1 (Predicted value). In a regression model $p(y|\tilde{x})$, the **predicted value** of y given $\tilde{x}$ is the estimated mean of Y given $\tilde{X} = \tilde{x}$:

$$\hat{y} \stackrel{\text{def}}{=} \hat{\mathbb{E}}[Y|\tilde{X} = \tilde{x}]$$

For linear models, the predicted value can be straightforwardly calculated by multiplying each predictor value x_j by its corresponding coefficient β_j and adding up the results:

$$\begin{aligned}\hat{y} &= \hat{\mathbb{E}}[Y|\tilde{X} = \tilde{x}] \\ &= \tilde{x}'\hat{\beta} \\ &= \hat{\beta}_0 \cdot 1 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p\end{aligned}$$

6.2 Example: prediction for the birthweight data

```
x <- c(1, 1, 40)
sum(x * coef(bw_lm1))
#> [1] 3225.49
```

R has built-in functions for prediction:

```
x <- tibble(age = 40, sex = "male")
bw_lm1 |> predict(newdata = x)
#>      1
#> 3225.49
```

If you don't provide `newdata`, R will use the covariate values from the original dataset:

```
predict(bw_lm1)
#>      1      2      3      4      5      6      7      8      9     10
#> 3225.49 3062.45 2983.70 2578.87 3225.49 3062.45 2621.02 2820.66 2741.91 3304.24
#>     11     12     13     14     15     16     17     18     19     20
#> 2862.81 2941.56 3346.38 3062.45 3225.49 2699.77 2862.81 2578.87 2983.70 2820.66
#>     21     22     23     24
#> 3225.49 2941.56 2983.70 3062.45
```

These special predictions are called the *fitted values* of the dataset:

Definition 6.2. For a given dataset $(\tilde{Y}, \mathbf{X})$ and corresponding fitted model $p_{\hat{\beta}}(\tilde{y}|\mathbf{x})$, the **fitted value** of y_i is the predicted value of y when $\tilde{X} = \tilde{x}_i$ using the estimate parameters $\hat{\beta}$.

R has an extra function to get these values:

```
fitted(bw_lm1)
#>      1      2      3      4      5      6      7      8      9     10
#> 3225.49 3062.45 2983.70 2578.87 3225.49 3062.45 2621.02 2820.66 2741.91 3304.24
#>     11     12     13     14     15     16     17     18     19     20
#> 2862.81 2941.56 3346.38 3062.45 3225.49 2699.77 2862.81 2578.87 2983.70 2820.66
#>     21     22     23     24
#> 3225.49 2941.56 2983.70 3062.45
```

6.3 Confidence intervals

Use `predict(se.fit = TRUE)` to compute SEs for predicted values:

```
bw_lm1 |>
  predict(
    newdata = x,
    se.fit = TRUE
  )
#> $fit
#>      1
#> 3225.49
#>
#> $se.fit
#> [1] 61.4599
#>
#> $df
#> [1] 21
#>
#> $residual.scale
#> [1] 177.116
```

The output of `predict.lm(se.fit = TRUE)` is a `list()`; you can extract the elements with `$` or `magrittr::use_series()`:

```
library(magrittr)
bw_lm1 |>
  predict(
    newdata = x,
    se.fit = TRUE
  ) |>
  use_series(se.fit)
#> [1] 61.4599
```

We can construct **confidence intervals** for $E[Y|X = x]$ using the usual formula:

$$\mu(\tilde{x}) \in (\hat{\mu}(\tilde{x}) \pm \zeta_{\alpha})$$

$$\zeta_{\alpha} = t_{n-p} \left(1 - \frac{\alpha}{2} \right) * \widehat{\text{se}}(\hat{\mu}(\tilde{x}))$$

$$\hat{\mu}(\tilde{x}) = \tilde{x} \cdot \hat{\beta}$$

$$\text{se}(\hat{\mu}(\tilde{x})) = \sqrt{\text{Var}(\hat{\mu}(\tilde{x}))}$$

$$\begin{aligned}
\text{Var}(\hat{\mu}(\tilde{x})) &= \text{Var}(x' \hat{\beta}) \\
&= x' \text{Var}(\hat{\beta}) x \\
&= x' \sigma^2 (\mathbf{X}' \mathbf{X})^{-1} x \\
&= \sigma^2 x' (\mathbf{X}' \mathbf{X})^{-1} x \\
&= \sum_{i=1}^n \sum_{j=1}^n x_i x_j \text{Cov}(\hat{\beta}_i, \hat{\beta}_j) \\
\widehat{\text{Var}}(\hat{\mu}(\tilde{x})) &= \hat{\sigma}^2 x' (\mathbf{X}' \mathbf{X})^{-1} x
\end{aligned}$$

```
bw_lm2 |> predict(
  newdata = x,
  interval = "confidence"
)
#>      fit      lwr      upr
#> 1 3210.64 3062.23 3359.05
```

```
library(sjPlot)
bw_lm2 |>
  plot_model(type = "pred", terms = c("age", "sex"), show.data = TRUE) +
  theme_sjplot() +
  theme(legend.position = "bottom")
```

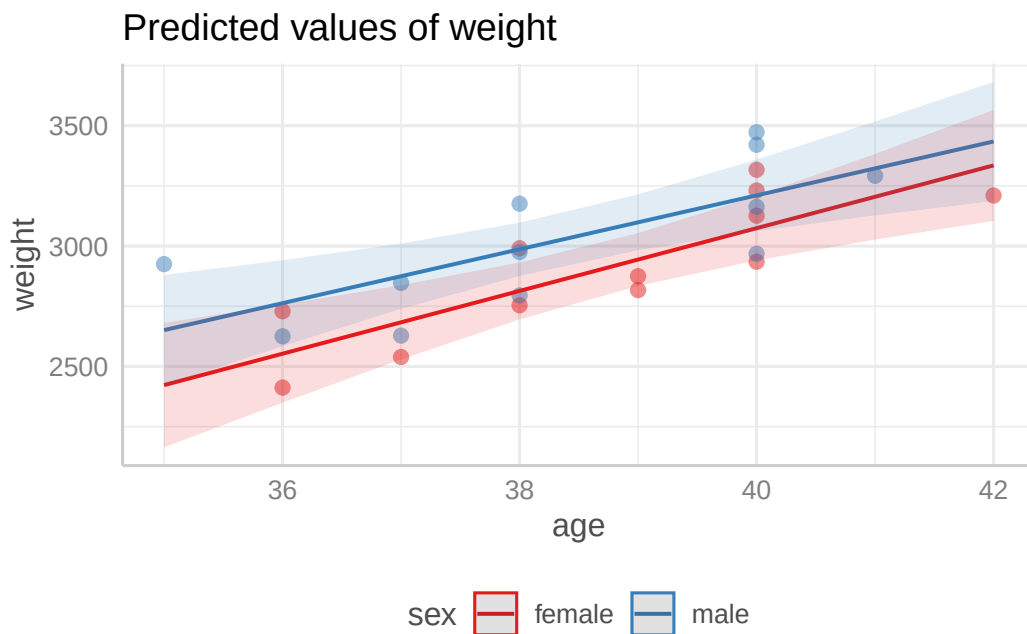


Figure 9: Predicted values and confidence bands for the `birthweight` model with interaction term

6.4 Prediction intervals

We can also construct **prediction intervals** for the value of a new observation Y^* , given a covariate pattern $\tilde{x}^*$:

$$\begin{aligned}
\hat{Y}^* &= \hat{\mu}(\tilde{x}^*) + \hat{\epsilon}^* \\
\text{Var}(\hat{Y}) &= \text{Var}(\hat{\mu}) + \text{Var}(\hat{\epsilon}) \\
\text{Var}(\hat{Y}) &= \text{Var}(x' \hat{\beta}) + \text{Var}(\hat{\epsilon}) \\
&= \tilde{x}^{*\top} \text{Var}(\hat{\beta}) \tilde{x}^* + \sigma^2 \\
&= \tilde{x}^{*\top} (\sigma^2 (\mathbf{X}'\mathbf{X})^{-1}) \tilde{x}^* + \sigma^2 \\
&= \sigma^2 \tilde{x}^{*\top} (\mathbf{X}'\mathbf{X})^{-1} \tilde{x}^* + \sigma^2
\end{aligned} \tag{10}$$

See Hogg et al. (2015) §7.6 (p. 340) for a longer version.

```
bw_lm2 |>
  predict(newdata = x, interval = "predict")
#>      fit      lwr      upr
#> 1 3210.64 2805.71 3615.57
```

If you don't specify `newdata`, you get a warning:

```
bw_lm2 |>
  predict(interval = "predict") |>
  head()
#> Warning in predict.lm(bw_lm2, interval = "predict"): predictions on current data refer to _future_
#>      fit      lwr      upr
#> 1 2552.73 2124.50 2980.97
#> 2 2552.73 2124.50 2980.97
#> 3 2683.13 2275.99 3090.27
#> 4 2813.53 2418.60 3208.47
#> 5 2813.53 2418.60 3208.47
#> 6 2943.93 2551.48 3336.38
```

The warning from the last command is: “predictions on current data refer to *future* responses” (since you already know what happened to the current data, and thus don't need to predict it).

See `?predict.lm` for more.

```
plot_PIs_and_CIs(bw_lm2, bw)
```

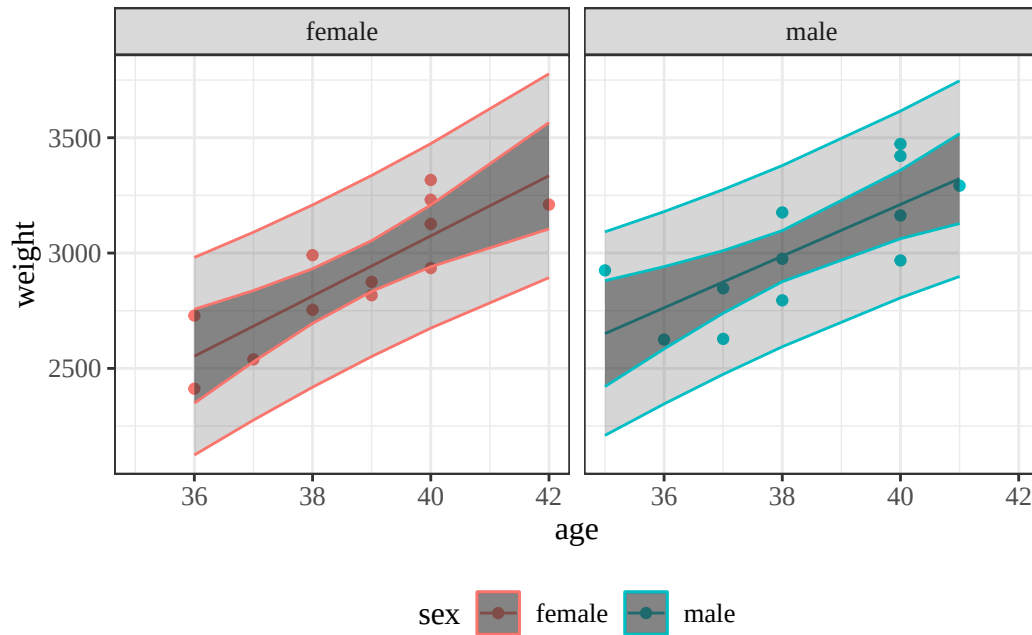


Figure 10: Confidence and prediction intervals for `birthweight` model 2

7 Diagnostics

💡 Tip

This section is adapted from Dobson and Barnett (2018, secs. 6.2–6.3) and Dunn and Smyth (2018) Chapter 3^a.

^ahttps://link.springer.com/chapter/10.1007/978-1-4419-0118-7_3

7.1 Assumptions in linear regression models

$$Y_i | \tilde{X}_i \sim \perp\!\!\!\perp N(\mu_i, \sigma^2)$$

$$\mu_i = \tilde{x} \cdot \tilde{\beta}$$

1. Normality

The model assumes that the distribution conditional on a given X value is Gaussian.

2. Correct Functional Form of Conditional Mean Structure (Linear Component)

The model assumes that the conditional means have the structure:

$$E[Y | \tilde{X} = \tilde{x}] = \tilde{x}' \tilde{\beta}$$

3. Homoskedasticity

The model assumes that variance σ^2 is constant (with respect to $\tilde{x}$).

4. Independence

The model assumes that the observations are statistically independent.

7.2 Direct visualization

The most direct way to examine the fit of a model is to compare it to the raw observed data.

```
bw <-  
  bw |>  
  mutate(  
    predlm2 = predict(bw_lm2)  
  ) |>  
  arrange(sex, age)  
  
plot1_interact <-  
  plot1 %>% bw +  
  geom_line(aes(y = predlm2))  
  
print(plot1_interact)
```

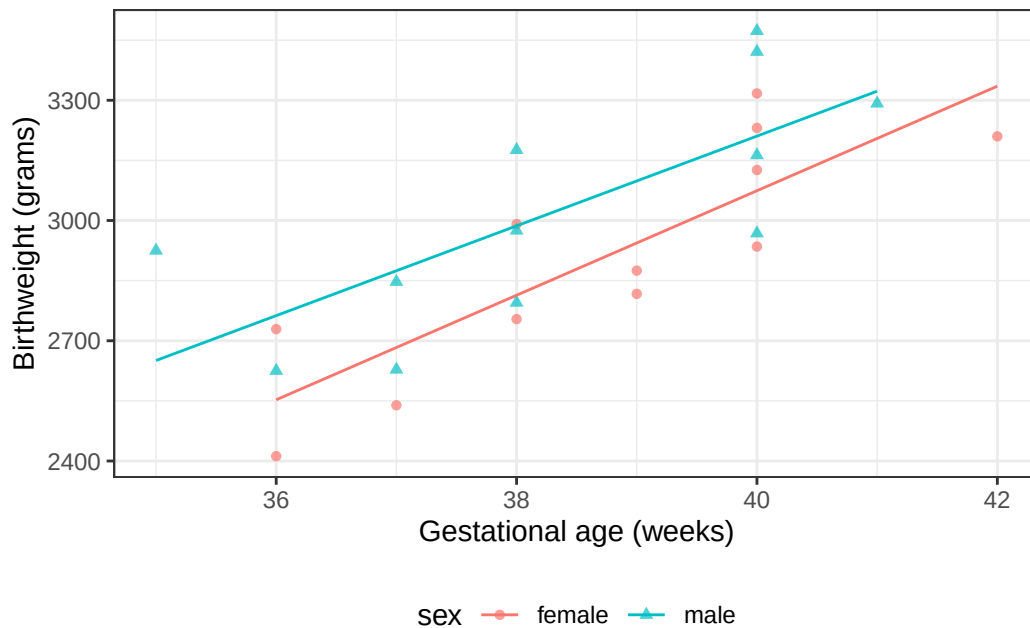


Figure 11: Birthweight model with interaction term

It's not easy to assess these assumptions from this model. If there are multiple continuous covariates, it becomes even harder to visualize the raw data.

7.2.1 Fitted model for hers data

Consider the `hers` data from Vittinghoff et al. (2012).

The “heart and estrogen/progestin study” (HERS) was a clinical trial of hormone therapy for prevention of recurrent heart attacks and death among 2,763 post-menopausal women with existing coronary heart disease (CHD) (Hulley et al. 1998).

Suppose we consider models with and without intercept terms (i.e., possibly forcing the intercept to go through 0):

```
hers_lm_with_int <- lm(  
  na.action = na.exclude,
```

Table 16: hers data

```

hers <- fs::path_package("rme", "extdata/hersdata.dta") |>
  haven::read_dta()
hers
#> # A tibble: 2,763 x 37
#>   HT          age raceth nonwhite smoking drinkany exercise physact globrat
#>   <dbl+lbl>   <dbl> <dbl+1> <dbl+1b> <dbl+1> <dbl+1b> <dbl+1b> <dbl+1> <dbl+1>
#> 1 0 [placebo]    70 2 [Afr~ 1 [yes] 0 [no] 0 [no] 0 [no] 5 [muc~ 3 [goo~
#> 2 0 [placebo]    62 2 [Afr~ 1 [yes] 0 [no] 0 [no] 0 [no] 1 [muc~ 3 [goo~
#> 3 1 [hormone ~    69 1 [Whi~ 0 [no] 0 [no] 0 [no] 0 [no] 3 [abo~ 3 [goo~
#> 4 0 [placebo]    64 1 [Whi~ 0 [no] 1 [yes] 1 [yes] 0 [no] 1 [muc~ 3 [goo~
#> 5 0 [placebo]    65 1 [Whi~ 0 [no] 0 [no] 0 [no] 0 [no] 2 [som~ 3 [goo~
#> 6 1 [hormone ~    68 2 [Afr~ 1 [yes] 0 [no] 1 [yes] 0 [no] 3 [abo~ 3 [goo~
#> 7 0 [placebo]    70 1 [Whi~ 0 [no] 0 [no] 0 [no] 0 [no] 3 [abo~ 2 [fai~
#> 8 1 [hormone ~    69 1 [Whi~ 0 [no] 0 [no] 0 [no] 1 [yes] 5 [muc~ 4 [ver~
#> 9 1 [hormone ~    61 1 [Whi~ 0 [no] 0 [no] 1 [yes] 1 [yes] 3 [abo~ 4 [ver~
#> 10 1 [hormone ~    62 1 [Whi~ 0 [no] 1 [yes] 1 [yes] 0 [no] 2 [som~ 3 [goo~
#> # i 2,753 more rows
#> # i 28 more variables: poorfair <dbl+lbl>, medcond <dbl>, htmeds <dbl+lbl>,
#> #   statins <dbl+lbl>, diabetes <dbl+lbl>, dmpills <dbl+lbl>,
#> #   insulin <dbl+lbl>, weight <dbl>, BMI <dbl>, waist <dbl>, WHR <dbl>,
#> #   glucose <dbl>, weight1 <dbl>, BMI1 <dbl>, waist1 <dbl>, WHR1 <dbl>,
#> #   glucose1 <dbl>, tchol <dbl>, LDL <dbl>, HDL <dbl>, TG <dbl>, tchol1 <dbl>,
#> #   LDL1 <dbl>, HDL1 <dbl>, TG1 <dbl>, SBP <dbl>, DBP <dbl>, age10 <dbl>

```

```

LDL ~ smoking * age, data = hers
)

library(equatiomatic)
equatiomatic::extract_eq(hers_lm_with_int)

```

$$LDL = \alpha + \beta_1(\text{smoking}) + \beta_2(\text{age}) + \beta_3(\text{smoking} \times \text{age}) + \epsilon \quad (11)$$

```

hers_lm_no_int <- lm(
  na.action = na.exclude,
  LDL ~ age + smoking:age - 1, data = hers
)

library(equatiomatic)
equatiomatic::extract_eq(hers_lm_no_int)

```

$$LDL = \beta_1(\text{age}) + \beta_2(\text{age} \times \text{age}_{\text{smoking}}) + \epsilon \quad (12)$$

Table 17: `hers` data models with and without intercepts

(a) With intercept

(b) No intercept

```
library(gtsummary)
hers_lm_with_int |>
  tbl_regression(intercept = TRUE)
```

```
hers_lm_no_int |>
  tbl_regression(intercept = TRUE)
```

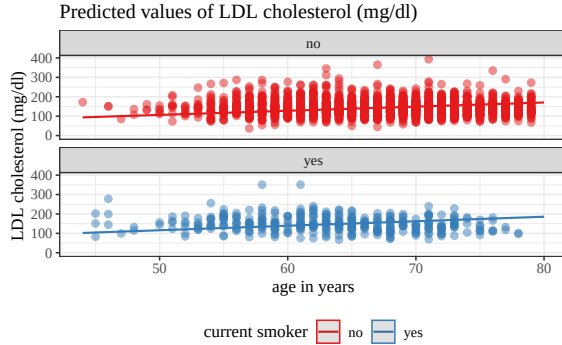
Characteristic	Beta	95% CI	p-value	Beta	95% CI	p-value
(Intercept)	154	138, 170	<0.001	2.1	2.1, 2.2	<0.001
current smoker	54	15, 94	0.007	0.19	0.12, 0.26	<0.001
age in years	-0.14	-0.38, 0.09	0.2			
current smoker * age in years	-0.79	-1.4, -0.17	0.012			

Abbreviation: CI = Confidence Interval

```
library(sjPlot)

hers_plot1 <- hers_lm_no_int |>
  sjPlot::plot_model(
    type = "pred",
    terms = c("age", "smoking"),
    show.data = TRUE
  ) +
  facet_wrap(~group_col, ncol = 1) +
  expand_limits(y = 0) +
  theme(legend.position = "bottom")
```

hers_plot1

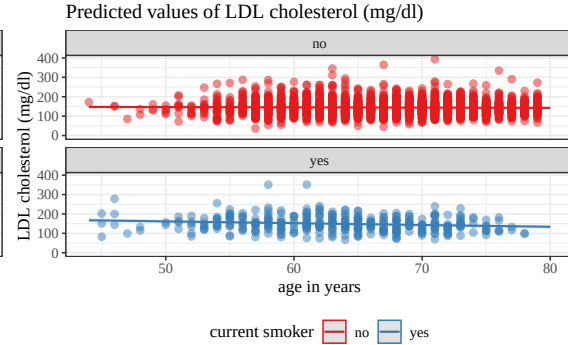


(a) No intercept

```
library(sjPlot)

hers_plot2 <- hers_lm_with_int |>
  sjPlot::plot_model(
    type = "pred",
    terms = c("age", "smoking"),
    show.data = TRUE
  ) +
  facet_wrap(~group_col, ncol = 1) +
  expand_limits(y = 0) +
  theme(legend.position = "bottom")
```

hers_plot2



(b) With intercept

Figure 12: `hers` data models with and without intercepts

7.3 Residuals

Maybe we can transform the data and model in some way to make it easier to inspect.

Definition 7.1 (Residual noise/deviation from the population mean). The **residual noise** in a probabilistic model $p(Y)$, also known as the **residual deviation of an observation from its population mean** or **residual** for short, is the difference between an observed value y and its population mean:

$$\varepsilon(y) \stackrel{\text{def}}{=} y - \mathbb{E}[Y] \quad (13)$$

We use the same notation for residual noise that we used for errors⁸.

$E[Y]$ can be viewed as an estimate of Y , before y is observed. Conversely, each observation y can be viewed as an estimate of $E[Y]$ (albeit an imprecise one, individually, since $n = 1$).

We can rearrange Equation 13 to view y as the sum of its mean plus the residual noise:

$$y = E[Y] + \varepsilon(y)$$

Theorem 7.1 (Residuals in Gaussian models). *If Y has a Gaussian distribution, then $\varepsilon(Y)$ also has a Gaussian distribution, and vice versa.*

Proof. Left to the reader. □

Definition 7.2 (Residuals of a fitted model value). The **residual of a fitted value** $\hat{y}$ (shorthand: “residual”) is its error⁹ relative to the observed data:

$$\begin{aligned} e(\hat{y}) &\stackrel{\text{def}}{=} \varepsilon(\hat{y}) \\ &= y - \hat{y} \end{aligned}$$

Example 7.1 (residuals in birthweight data).

⁸[estimation.qmd#def-error](#)

⁹[estimation.qmd#def-error](#)

```

plot1_interact +
  facet_wrap(~sex) +
  geom_segment(
    aes(
      x = age,
      y = predlm2,
      xend = age,
      yend = weight,
      col = sex,
      group = id
    ),
    linetype = "dotted"
  )

```

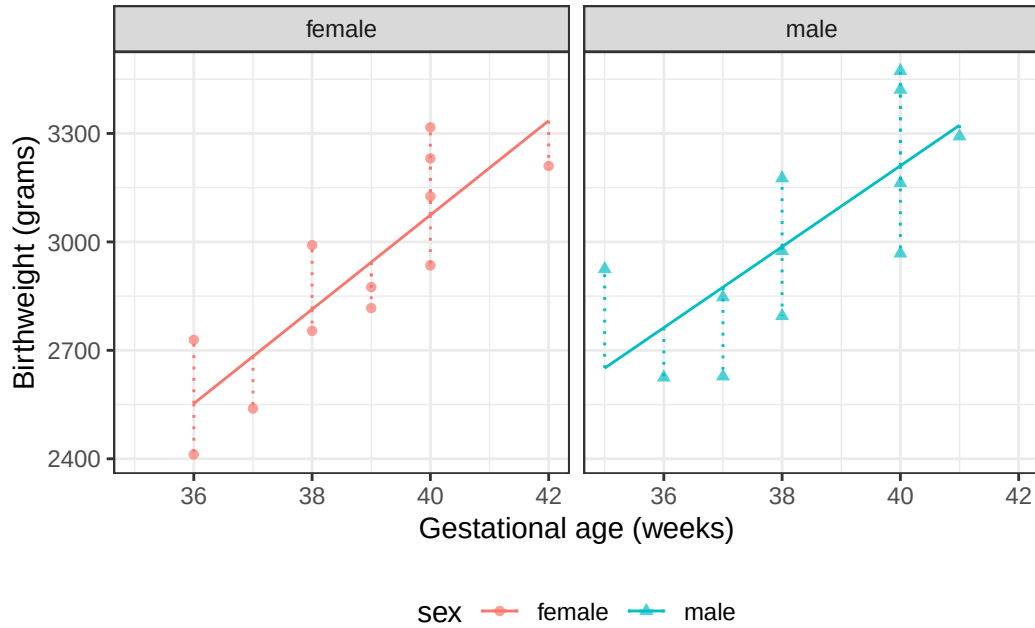


Figure 13: Fitted values and residuals for interaction model for `birthweight` data

7.3.1 Residuals of fitted values vs residual noise

$e(\hat{y})$ can be seen as the maximum likelihood estimate of the residual noise:

$$\begin{aligned}
 e(\hat{y}) &= y - \hat{y} \\
 &= \hat{\varepsilon}_{ML}
 \end{aligned}$$

7.3.2 General characteristics of residuals

Theorem 7.2. If $\hat{E}[Y]$ is an unbiased¹⁰ estimator of the mean $E[Y]$, then:

$$E[e(y)] = 0 \quad (14)$$

$$\text{Var}(e(y)) \approx \sigma^2 \quad (15)$$

¹⁰[estimation.qmd#sec-unbiased-estimators](#)

Proof.

Equation 14:

$$\begin{aligned} E[e(y)] &= E[y - \hat{y}] \\ &= E[y] - E[\hat{y}] \\ &= E[y] - E[y] \\ &= 0 \end{aligned}$$

Equation 15:

$$\begin{aligned} \text{Var}(e(y)) &= \text{Var}(y - \hat{y}) \\ &= \text{Var}(y) + \text{Var}(\hat{y}) - 2\text{Cov}(y, \hat{y}) \\ &\approx \text{Var}(y) + 0 - 2 \cdot 0 \\ &= \text{Var}(y) \\ &= \sigma^2 \end{aligned}$$

□

7.3.3 Characteristics of residuals in Gaussian models

With enough data and a correct model, the residuals will be approximately Gaussian distributed, with variance σ^2 , which we can estimate using $\hat{\sigma}^2$; that is:

$$e_i \sim_{\text{iid}} N(0, \hat{\sigma}^2)$$

7.3.4 Computing residuals in R

R provides a function for residuals:

```
resid(bw_lm2)
#>      1      2      3      4      5      6      7      8
#> 176.2667 -140.7333 -144.1333 -59.5333 177.4667 -126.9333 -68.9333 242.6667
#>      9     10     11     12     13     14     15     16
#> -139.3333  51.6667 156.6667 -125.1333 274.2759 -137.7069 -27.6897 -246.6897
#>     17     18     19     20     21     22     23     24
#> -191.6724 189.3276 -11.6724 -242.6379 -47.6379 262.3621 210.3621 -30.6207
```

Exercise 7.1. Check R's output by computing the residuals directly.

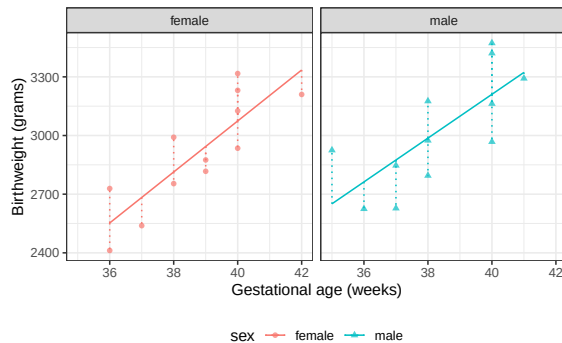
Solution.

```
bw$weight - fitted(bw_lm2)
#>      1      2      3      4      5      6      7      8
#> 176.2667 -140.7333 -144.1333 -59.5333 177.4667 -126.9333 -68.9333 242.6667
#>      9     10     11     12     13     14     15     16
#> -139.3333  51.6667 156.6667 -125.1333 274.2759 -137.7069 -27.6897 -246.6897
#>     17     18     19     20     21     22     23     24
#> -191.6724 189.3276 -11.6724 -242.6379 -47.6379 262.3621 210.3621 -30.6207
```

This matches R's output!

7.3.5 Graphing the residuals

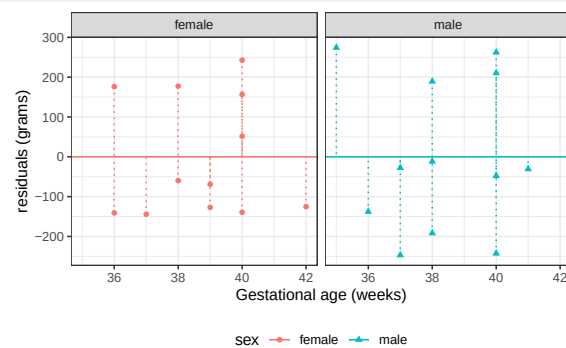
```
plot1_interact +
  facet_wrap(~sex) +
  geom_segment(
    aes(
      x = age,
      y = predlm2,
      xend = age,
      yend = weight,
      col = sex,
      group = id
    ),
    linetype = "dotted"
  )
```



(a) fitted values

```
bw <- bw |>
  mutate(
    resids_intxn =
      weight - fitted(bw_lm2)
  )

plot_bw_resid <-
  bw |>
  ggplot(aes(
    x = age,
    y = resids_intxn,
    linetype = sex,
    shape = sex,
    col = sex
  )) +
  theme_bw() +
  xlab("Gestational age (weeks)") +
  ylab("residuals (grams)") +
  theme(legend.position = "bottom") +
  geom_hline(aes(
    yintercept = 0,
    col = sex
  )) +
  geom_segment(
    aes(yend = 0),
    linetype = "dotted"
  ) +
  geom_point()
# expand_limits(y = 0, x = 0) +
# geom_point(alpha = .7)
#> geom_point: na.rm = FALSE
#> stat_identity: na.rm = FALSE
#> position_identity
print(plot_bw_resid + facet_wrap(~sex))
```



(b) Residuals

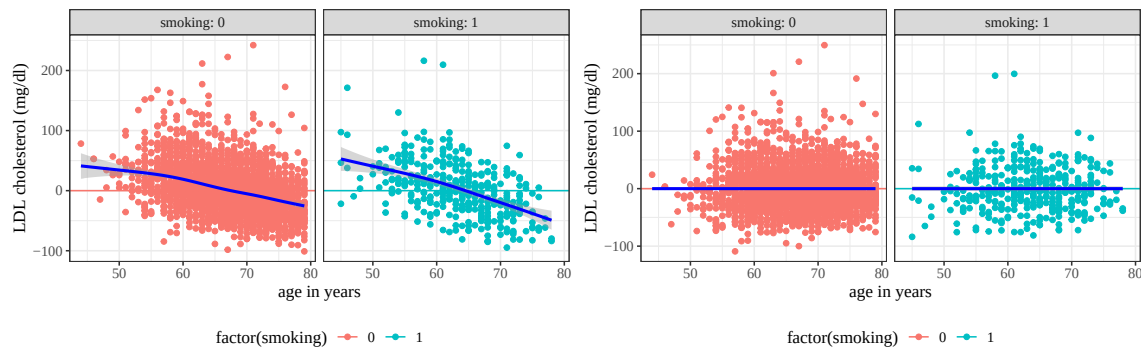
Figure 14: Fitted values and residuals for interaction model for `birthweight` data

7.3.6 Residuals versus predictors

```
hers <- hers |>
  mutate(
    resids_no_intcpt =
      LDL - fitted(hers_lm_no_int),
    resids_with_intcpt =
      LDL - fitted(hers_lm_with_int)
  )
```

```
hers |>
  arrange(age) |>
  ggplot() +
    aes(x = age, y = resids_no_intcpt, col = factor(smoking)) +
    geom_point() +
    geom_hline(aes(yintercept = 0, col = factor(smoking))) +
    facet_wrap(~smoking, labeller = "label_both") +
    theme(legend.position = "bottom") +
    geom_smooth(col = "blue")
```

```
hers |>
  arrange(age) |>
  ggplot() +
    aes(x = age, y = resids_with_intcpt, col = factor(smoking)) +
    geom_point() +
    geom_hline(aes(yintercept = 0, col = factor(smoking))) +
    facet_wrap(~smoking, labeller = "label_both") +
    theme(legend.position = "bottom") +
    geom_smooth(col = "blue")
```



(a) no intercept

(b) with intercept

Figure 15: Residuals of `hers` data vs predictors

7.3.7 Residuals versus fitted values

If the model contains multiple continuous covariates, how do we check for errors in the mean structure assumption?

```
library(ggfortify)
hers_lm_no_int |>
  update(na.action = na.omit) |>
  autoplot(
    which = 1,
    ncol = 1,
    smooth.colour = NA
  ) +
  geom_hline(yintercept = 0, col = "red")
```

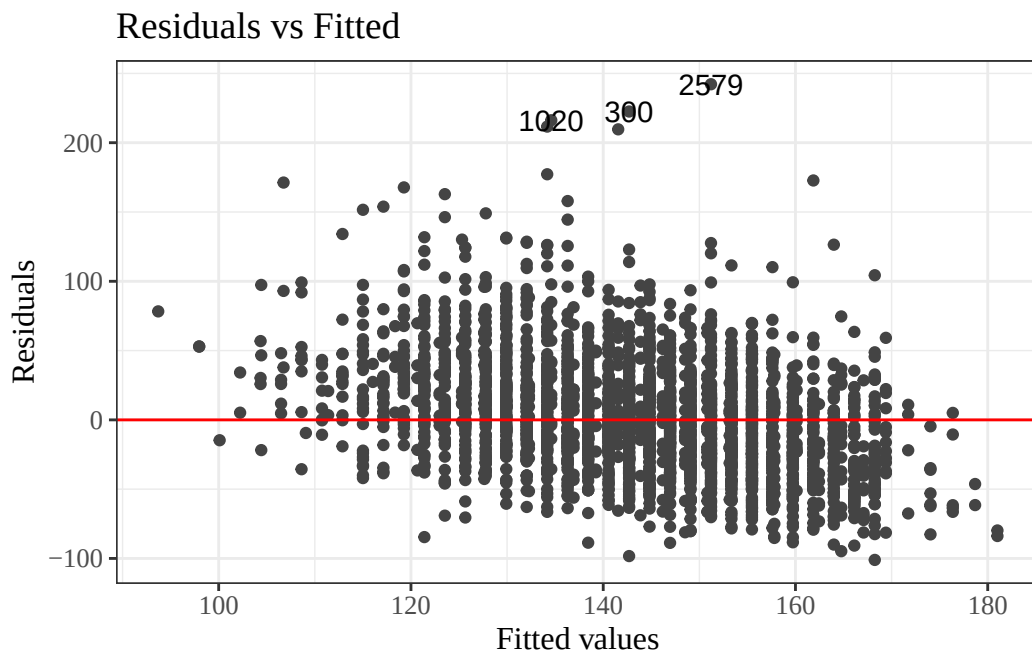


Figure 16: Residuals of interaction model for `hers` data

We can add a LOESS smooth to visualize where the residual mean is nonzero:

```
library(ggfortify)
hers_lm_no_int |>
  update(na.action = na.omit) |>
  autoplot(
    which = 1,
    ncol = 1
  ) +
  geom_hline(yintercept = 0, col = "red")
```

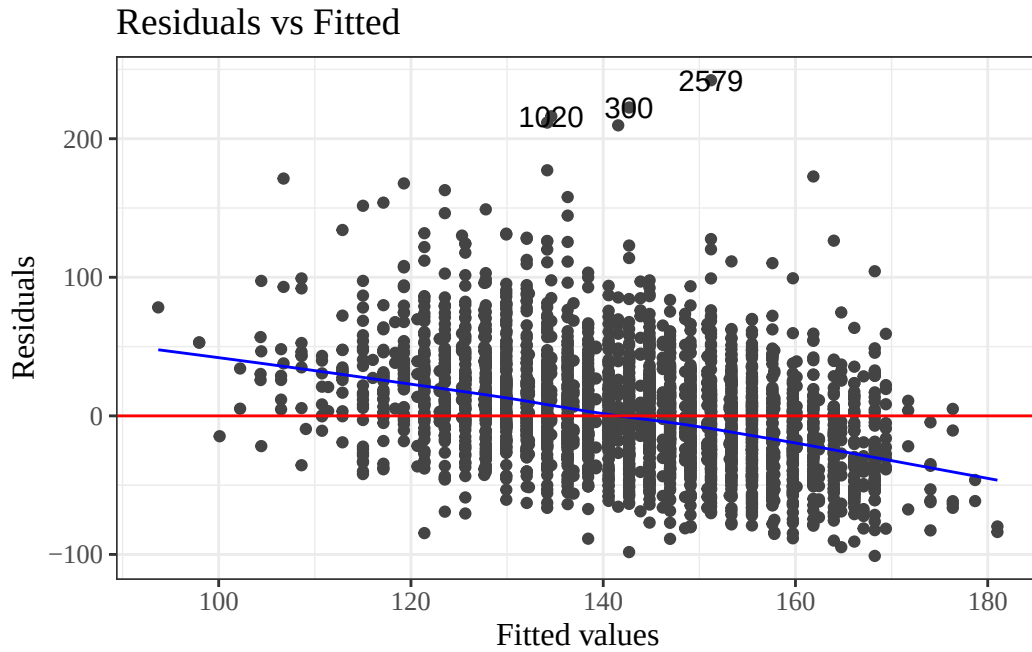


Figure 17: Residuals of interaction model for `hers` data, no intercept term

```
library(ggfortify)
hers_lm_no_int |>
  update(na.action = na.omit) |>
  autoplot(
    which = 1,
    ncol = 1
  ) +
  geom_hline(yintercept = 0, col = "red")

hers_lm_with_int |>
  update(na.action = na.omit) |>
  autoplot(
    which = 1,
    ncol = 1
  ) +
  geom_hline(yintercept = 0, col = "red")
```

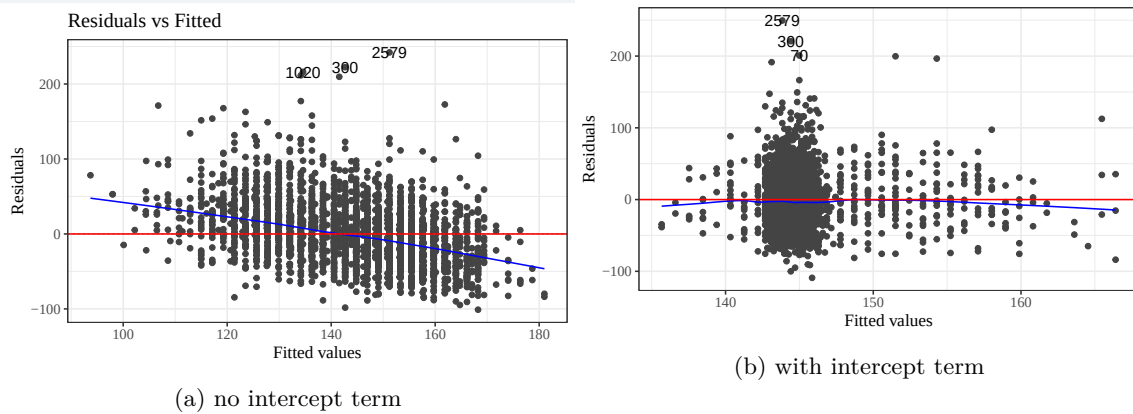


Figure 18: Residuals of interaction model for `hers` data, with and without intercept term

Definition 7.3 (Standardized residuals).

$$r_i = \frac{e_i}{\widehat{SD}(e_i)}$$

Hence, with enough data and a correct model, the standardized residuals will be approximately standard Gaussian; that is,

$$r_i \sim_{\text{iid}} N(0, 1)$$

7.4 Marginal distributions of residuals

To look for problems with our model, we can check whether the residuals e_i and standardized residuals r_i look like they have the distributions that they are supposed to have, according to the model.

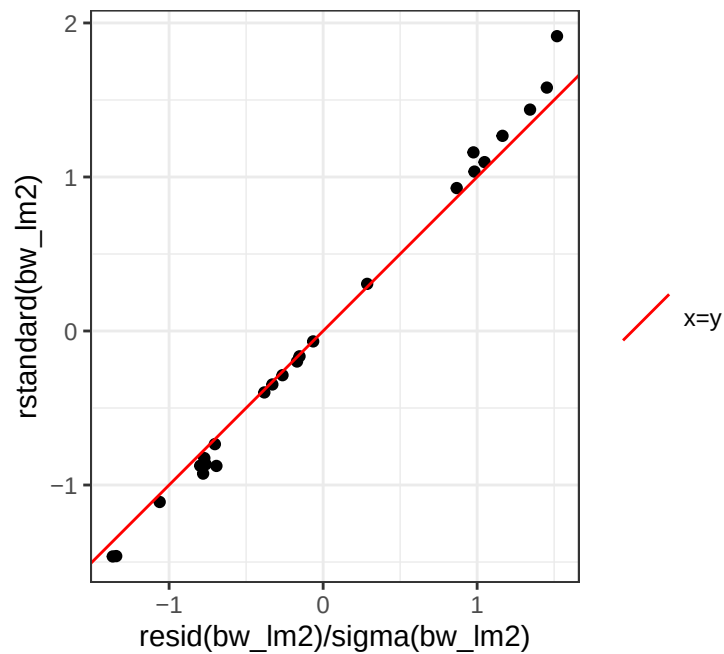
7.4.1 Standardized residuals in R

```
rstandard(bw_lm2)
#>      1      2      3      4      5      6      7
#>  1.1598166 -0.9260109 -0.8747917 -0.3472255  1.0350665 -0.7347315 -0.3990086
#>      8      9     10     11     12     13     14
#>  1.4375164 -0.8253872  0.3060646  0.9280669 -0.8761592  1.9142780 -0.8655921
#>     15     16     17     18     19     20     21
#> -0.1642993 -1.4637574 -1.1101599  1.0965787 -0.0676062 -1.4615865 -0.2869582
#>     22     23     24
#>  1.5803994  1.2671652 -0.1980543
resid(bw_lm2) / sigma(bw_lm2)
#>      1      2      3      4      5      6      7
#>  0.9759331 -0.7791962 -0.7980209 -0.3296173  0.9825771 -0.7027900 -0.3816622
#>      8      9     10     11     12     13     14
#>  1.3435690 -0.7714449  0.2860621  0.8674141 -0.6928239  1.5185792 -0.7624398
#>     15     16     17     18     19     20     21
#> -0.1533089 -1.3658431 -1.0612299  1.0482473 -0.0646265 -1.3434099 -0.2637562
#>     22     23     24
#>  1.4526163  1.1647086 -0.1695371
```

These are not quite the same, because R is doing something more complicated and precise to get the standard errors. Let's not worry about those details for now; the difference is pretty small in this case:

```
rstandard_compare_plot <-
  tibble(
    x = resid(bw_lm2) / sigma(bw_lm2),
    y = rstandard(bw_lm2)
  ) |>
  ggplot(aes(x = x, y = y)) +
  geom_point() +
  theme_bw() +
  coord_equal() +
  xlab("resid(bw_lm2)/sigma(bw_lm2)") +
  ylab("rstandard(bw_lm2)") +
  geom_abline(
    aes(
      intercept = 0,
      slope = 1,
      col = "x=y"
    )
  ) +
  labs(colour = "") +
  scale_colour_manual(values = "red")

print(rstandard_compare_plot)
```



Let's add these residuals to the tibble of our dataset:

```
bw <-
  bw |>
  mutate(
    fitted_lm2 = fitted(bw_lm2),
    resid_lm2 = resid(bw_lm2),
    resid_lm2_alt = weight - fitted_lm2,
    std_resid_lm2 = rstandard(bw_lm2),
    std_resid_lm2_alt = resid_lm2 / sigma(bw_lm2)
  )

bw |>
  select(
    sex,
    age,
    weight,
    fitted_lm2,
    resid_lm2,
    std_resid_lm2
  )

#> # A tibble: 24 x 6
#>   sex      age weight fitted_lm2 resid_lm2 std_resid_lm2
#>   <fct> <dbl> <dbl>      <dbl>      <dbl>      <dbl>
#> 1 female   36  2729      2553.      176.         1.16
#> 2 female   36  2412      2553.     -141.        -0.926
#> 3 female   37  2539      2683.     -144.        -0.875
#> 4 female   38  2754      2814.     -59.5        -0.347
#> 5 female   38  2991      2814.      177.         1.04
#> 6 female   39  2817      2944.     -127.        -0.735
#> 7 female   39  2875      2944.     -68.9        -0.399
#> 8 female   40  3317      3074.      243.         1.44
#> 9 female   40  2935      3074.     -139.        -0.825
```

```
#> 10 female    40   3126    3074.    51.7    0.306
#> # i 14 more rows
```

Now let's build histograms:

```
resid_marginal_hist <-
  bw |>
  ggplot(aes(x = resid_lm2)) +
  geom_histogram()

print(resid_marginal_hist)
```

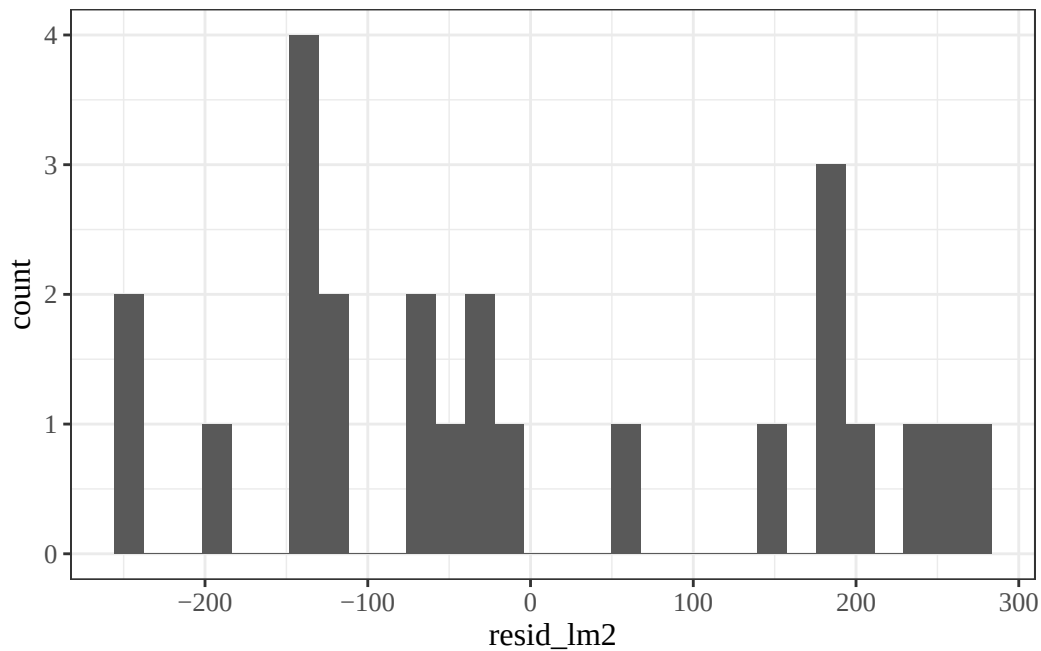


Figure 19: Marginal distribution of (nonstandardized) residuals

Hard to tell with this small amount of data, but I'm a bit concerned that the histogram doesn't show a bell-curve shape.

```
std_resid_marginal_hist <-
  bw |>
  ggplot(aes(x = std_resid_lm2)) +
  geom_histogram()

print(std_resid_marginal_hist)
```

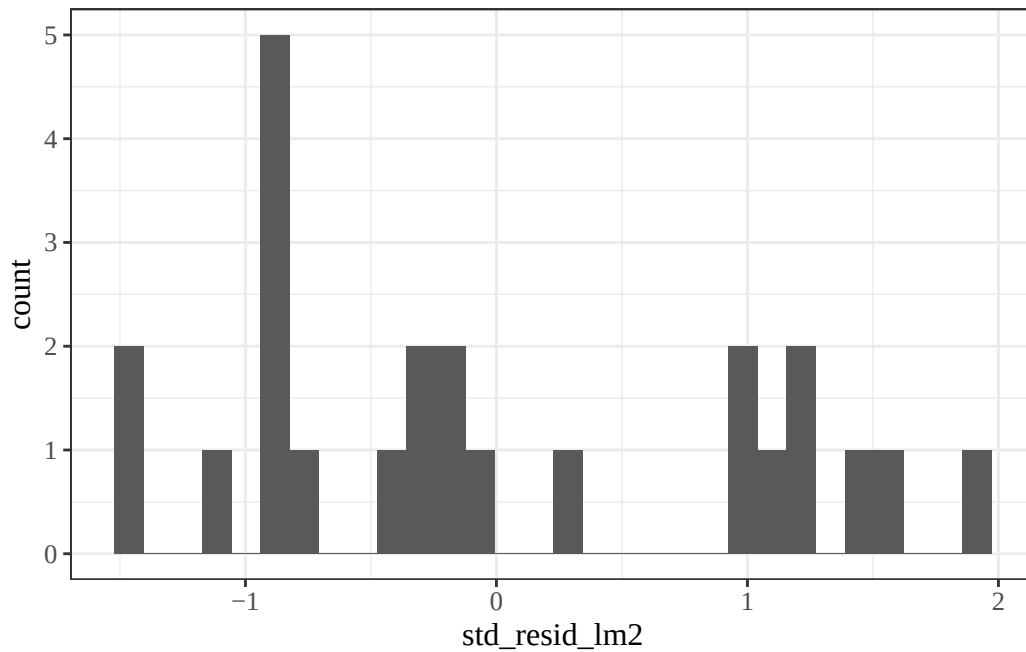


Figure 20: Marginal distribution of standardized residuals

This looks similar, although the scale of the x-axis got narrower, because we divided by $\hat{\sigma}$ (roughly speaking).

Still hard to tell if the distribution is Gaussian.

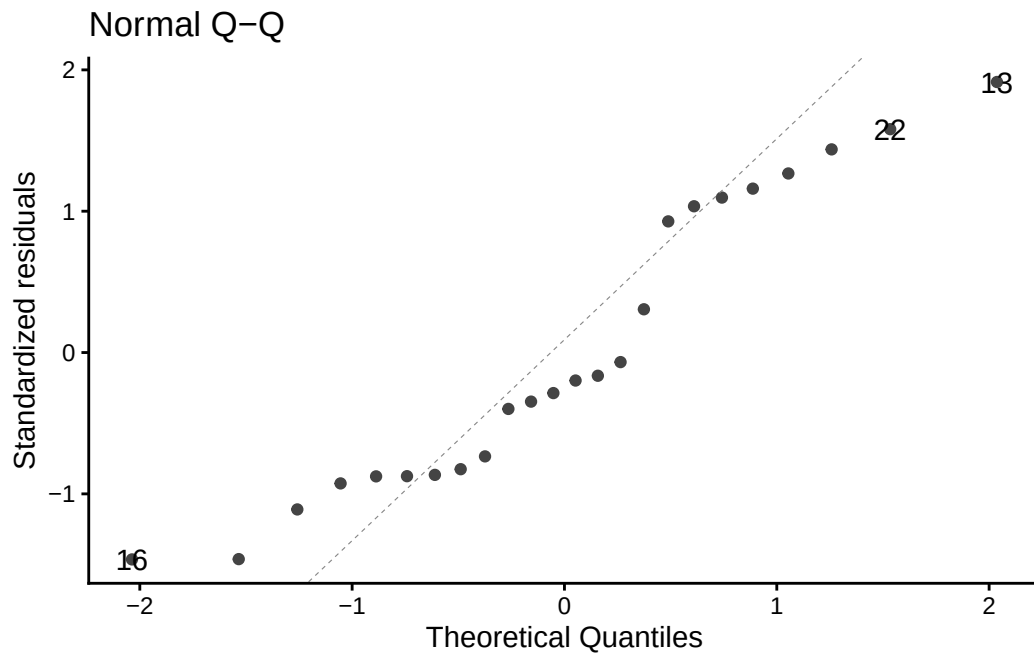
7.5 QQ plot of standardized residuals

Another way to assess normality is the QQ plot of the standardized residuals versus normal quantiles:

```
library(ggfortify)
# needed to make ggplot2::autoplot() work for `lm` objects

qqplot_lm2_auto <-
  bw_lm2 |>
  autoplot(
    which = 2, # options are 1:6; can do multiple at once
    ncol = 1
  ) +
  theme_classic()

print(qqplot_lm2_auto)
```



If the Gaussian model were correct, these points should follow the dotted line.

Fig 2.4 panel (c) in Dobson and Barnett (2018) is a little different; they didn't specify how they produced it, but other statistical analysis systems do things differently from R.

See also Dunn and Smyth (2018) §3.5.4¹¹.

7.5.1 QQ plot - how it's built

Let's construct it by hand:

```
bw <- bw |>
  mutate(
    p = (rank(std_resid_lm2) - 1 / 2) / n(), # "Blom's method"
    expected_quantiles_lm2 = qnorm(p)
  )

qqplot_lm2 <-
  bw |>
  ggplot(
    aes(
      x = expected_quantiles_lm2,
      y = std_resid_lm2,
      col = sex,
      shape = sex
    )
  ) +
  geom_point() +
  theme_classic() +
  theme(legend.position = "none") + # removing the plot legend
  ggtitle("Normal Q-Q") +
  xlab("Theoretical Quantiles") +
  ylab("Standardized residuals")
```

¹¹https://link.springer.com/chapter/10.1007/978-1-4419-0118-7_3#Sec14:~:text=3.5.4%20Q%E2%80%93Q%20Plots%20and%20Normality

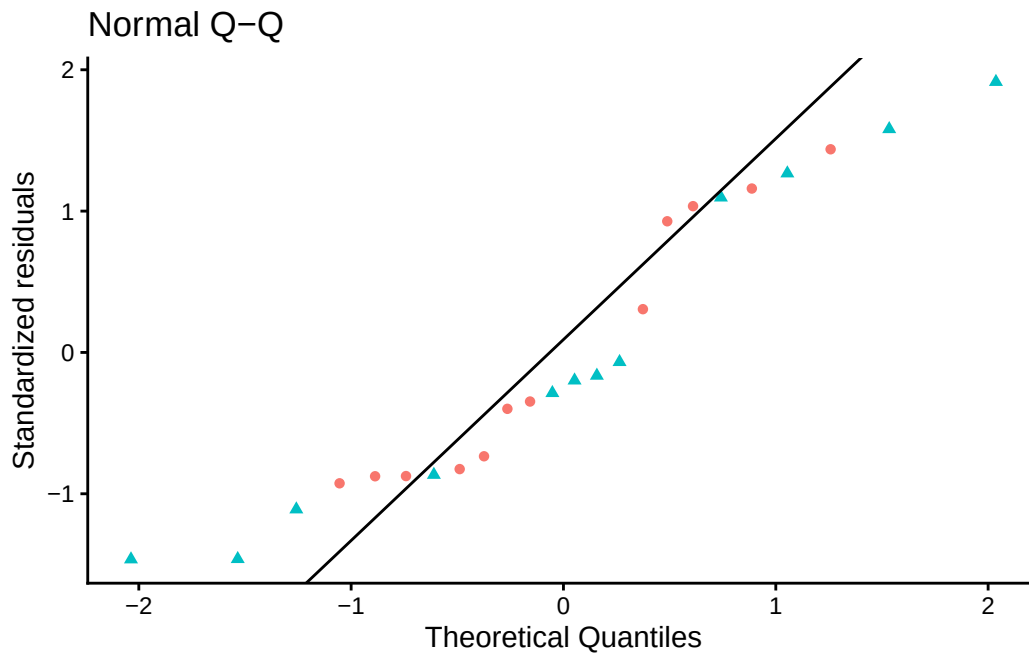

```
# find the expected line:

ps <- c(.25, .75) # reference probabilities
a <- quantile(rstandard(bw_lm2), ps) # empirical quantiles
b <- qnorm(ps) # theoretical quantiles

qq_slope <- diff(a) / diff(b)
qq_intcpt <- a[1] - b[1] * qq_slope

qqplot_lm2 <-
  qqplot_lm2 +
  geom_abline(slope = qq_slope, intercept = qq_intcpt)

print(qqplot_lm2)
```



7.6 Conditional distributions of residuals

If our Gaussian linear regression model is correct, the residuals e_i and standardized residuals r_i should have:

- an approximately Gaussian distribution, with:
- a mean of 0
- a constant variance

This should be true **for every** value of x .

If we didn't correctly guess the functional form of the linear component of the mean,

$$E[Y|X = x] = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

Then the residuals might have nonzero mean.

Regardless of whether we guessed the mean function correctly, the variance of the residuals might differ between values of x .

7.6.1 Residuals versus fitted values

To look for these issues, we can plot the residuals e_i against the fitted values $\hat{y}_i$ (Figure 21).

```
autoplot(bw_lm2, which = 1, ncol = 1) |> print()
```

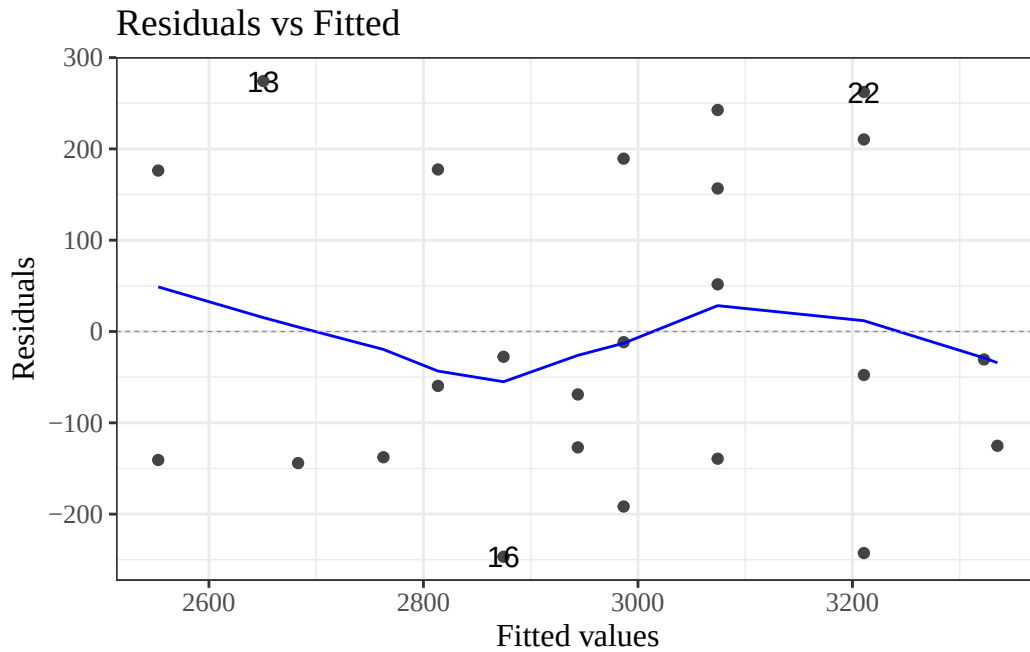


Figure 21: birthweight model (Equation 2): residuals versus fitted values

If the model is correct, the blue line should stay flat and close to 0, and the cloud of dots should have the same vertical spread regardless of the fitted value.

If not, we probably need to change the functional form of linear component of the mean,

$$E[Y|X = x] = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p$$

Example: PLOS Medicine title length data

(Adapted from Dobson and Barnett (2018), §6.7.1)

```
data(PLOS, package = "dobson")
library(ggplot2)
fig1 =
  PLOS |>
  ggplot(
    aes(x = authors,
        y = nchar)
  ) +
  geom_point() +
  theme(legend.position = "bottom") +
  labs(col = "") +
  guides(col=guide_legend(ncol=3))
fig1
```

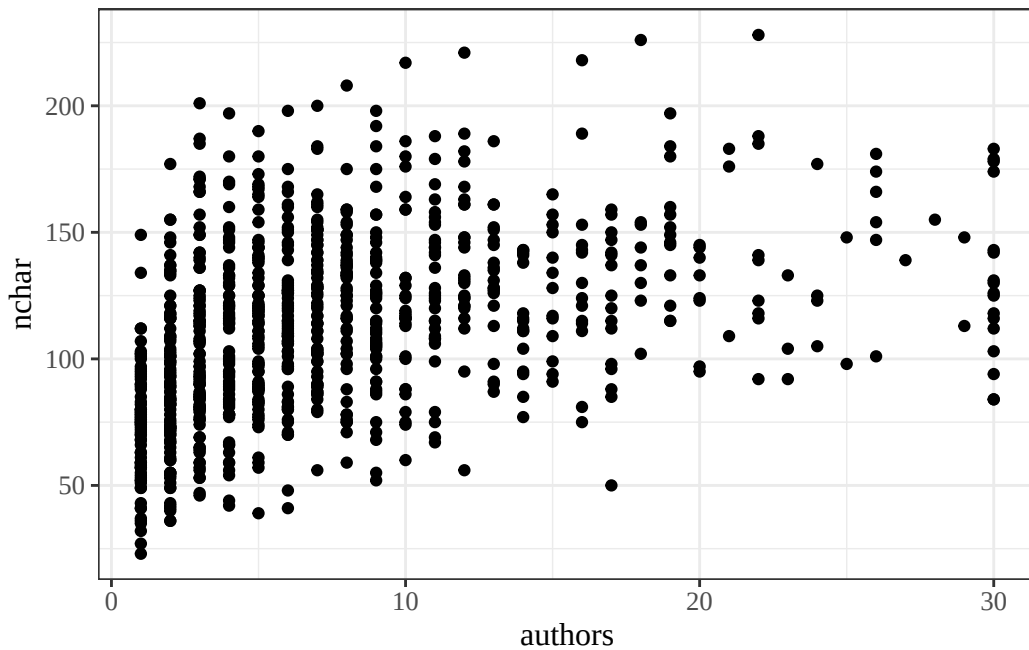


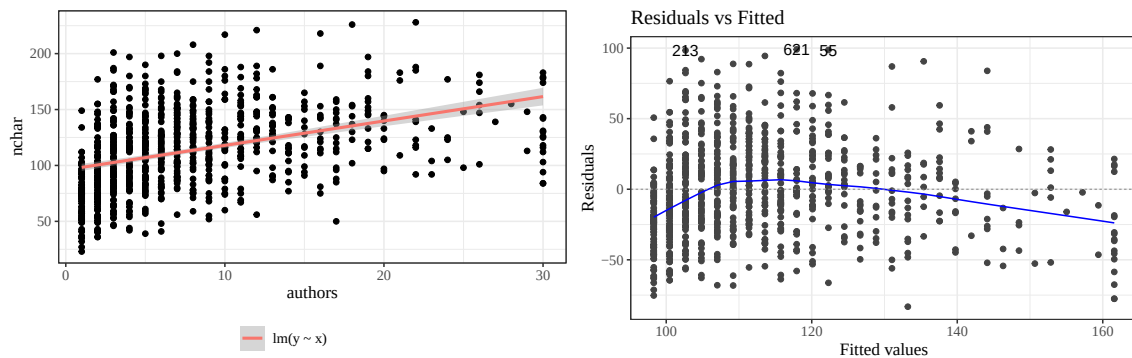
Figure 22: Number of authors versus title length in *PLOS Medicine* articles

Linear fit

```
lm_PLOS_linear = lm(
  formula = nchar ~ authors,
  data = PLOS)

fig2 = fig1 +
  geom_smooth(
    method = "lm",
    fullrange = TRUE,
    aes(col = "lm(y ~ x)"))
fig2

library(ggfortify)
autoplot(lm_PLOS_linear, which = 1, ncol = 1)
```



(a) Data and fit

(b) Residuals vs fitted

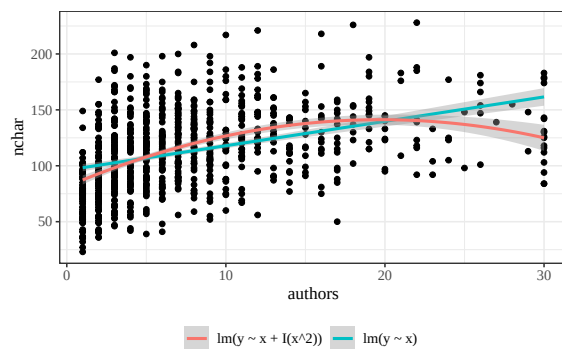
Figure 23: Number of authors versus title length in *PLOS Medicine*, with linear model fit

Quadratic fit

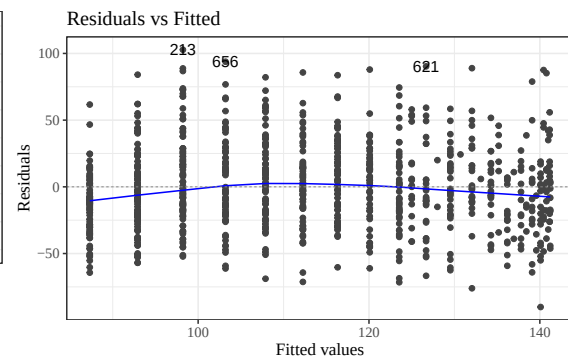
```
lm_PLOS_quad = lm(
  formula = nchar ~ authors + I(authors^2),
  data = PLOS)
```

```
fig3 =
  fig2 +
  geom_smooth(
    method = "lm",
    fullrange = TRUE,
    formula = y ~ x + I(x ^ 2),
    aes(col = "lm(y ~ x + I(x^2))")
  )
fig3

autoplot(lm_PLOS_quad, which = 1, ncol = 1)
```



(a) Data and fit



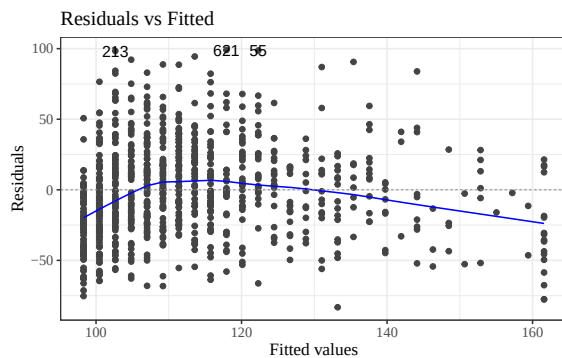
(b) Residuals vs fitted

Figure 24: Number of authors versus title length in *PLOS Medicine*, with quadratic model fit

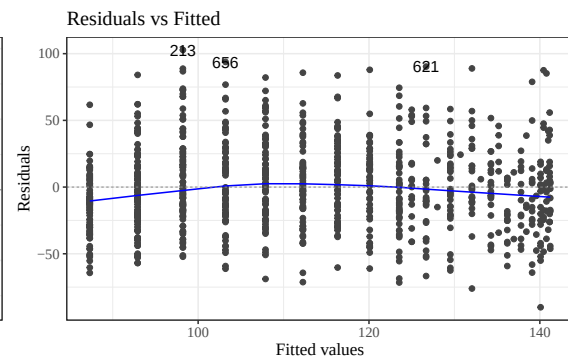
Linear versus quadratic fits

```
library(ggfortify)
autoplot(lm_PLOS_linear, which = 1, ncol = 1)

autoplot(lm_PLOS_quad, which = 1, ncol = 1)
```



(a) Linear



(b) Quadratic

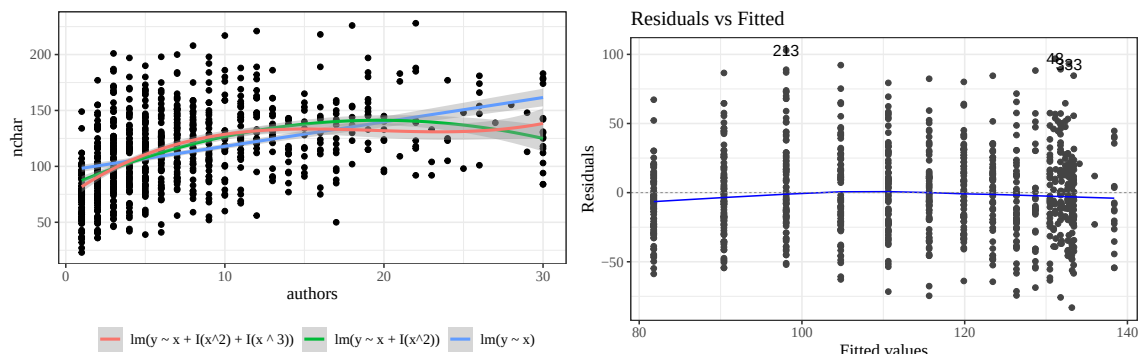
Figure 25: Residuals versus fitted plot for linear and quadratic fits to PLOS data

Cubic fit

```
lm_PLOS_cub = lm(
  formula = nchar ~ authors + I(authors^2) + I(authors^3),
  data = PLOS)
```

```
fig4 =
  fig3 +
  geom_smooth(
    method = "lm",
    fullrange = TRUE,
    formula = y ~ x + I(x ^ 2) + I(x ^ 3),
    aes(col = "lm(y ~ x + I(x^2) + I(x ^ 3))")
  )
fig4

autoplot(lm_PLOS_cub, which = 1, ncol = 1)
```



(a) Data and fit

(b) Residuals vs fitted

Figure 26: Number of authors versus title length in *PLOS Medicine*, with cubic model fit

Logarithmic fit

```
lm_PLOS_log = lm(nchar ~ log(authors), data = PLOS)
```

```
fig5 = fig4 +
  geom_smooth(
    method = "lm",
    fullrange = TRUE,
    formula = y ~ log(x),
    aes(col = "lm(y ~ log(x))")
  )
fig5

autoplot(lm_PLOS_log, which = 1, ncol = 1)
```

Table 18: linear vs quadratic

```
anova(lm_PLOS_linear, lm_PLOS_quad)
#> # A tibble: 2 x 6
#>   Res.Df    RSS    Df `Sum of Sq`      F `Pr(>F)`
#>   <dbl>   <dbl> <dbl>   <dbl> <dbl>   <dbl>
#> 1     876 947502.    NA        NA    NA    NA
#> 2     875 880950.     1    66552.  66.1 1.46e-15
```

Table 19: quadratic vs cubic

```
anova(lm_PLOS_quad, lm_PLOS_cub)
#> # A tibble: 2 x 6
#>   Res.Df    RSS    Df `Sum of Sq`      F `Pr(>F)`
#>   <dbl>   <dbl> <dbl>   <dbl> <dbl>   <dbl>
#> 1     875 880950.    NA        NA    NA    NA
#> 2     874 865933.     1    15018.  15.2 0.000106
```

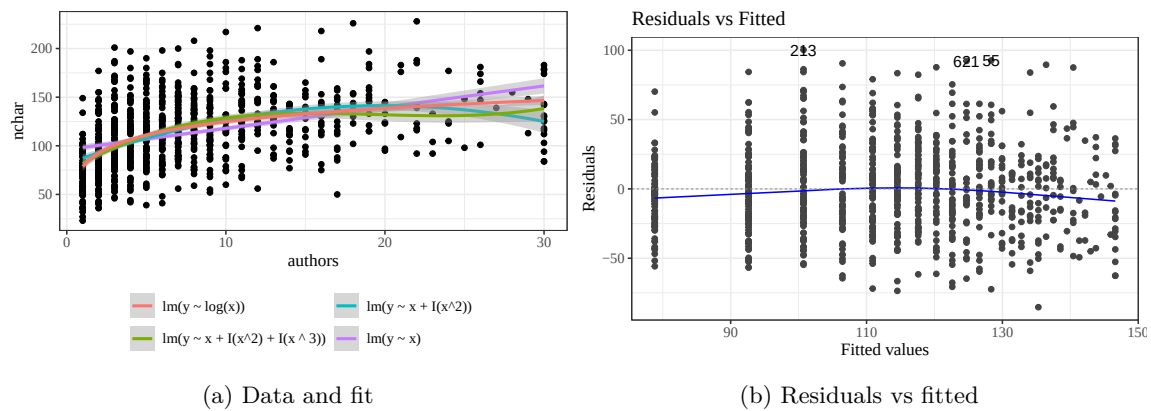


Figure 27: logarithmic fit

Model selection

AIC/BIC

```
AIC(lm_PLOS_quad)
#> [1] 8567.61
AIC(lm_PLOS_cub)
#> [1] 8554.51
```

```
AIC(lm_PLOS_cub)
#> [1] 8554.51
AIC(lm_PLOS_log)
#> [1] 8543.63
```

```
BIC(lm_PLOS_cub)
#> [1] 8578.4
BIC(lm_PLOS_log)
#> [1] 8557.97
```

Extrapolation is dangerous

```
fig_all = fig5 +  
  xlim(0, 60)  
fig_all
```

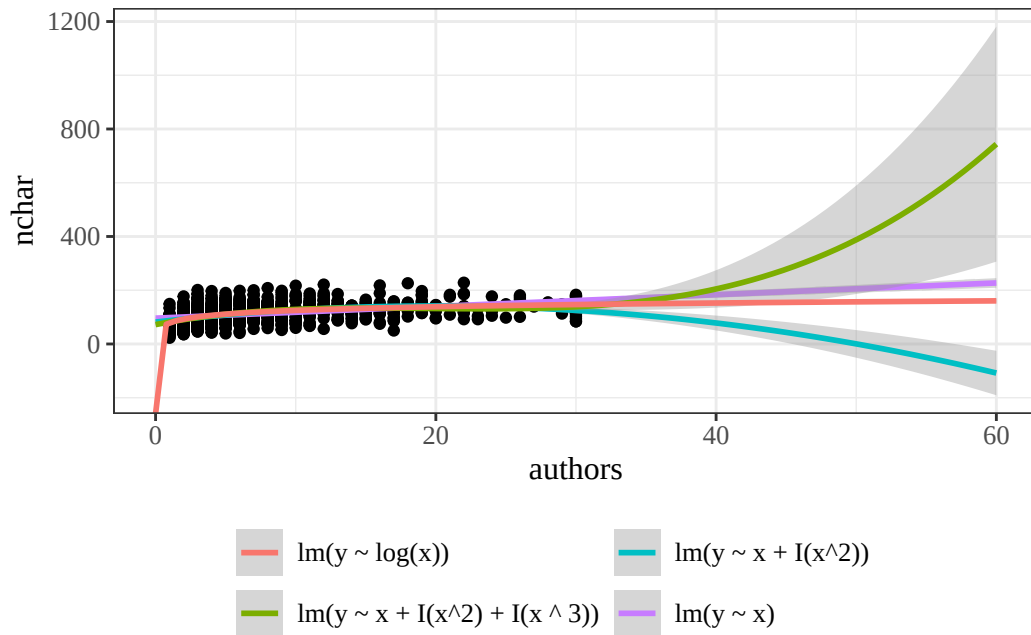


Figure 28: Number of authors versus title length in *PLOS Medicine*

7.6.2 Scale-location plot

We can also plot the square roots of the absolute values of the standardized residuals against the fitted values (Figure 29).

```
autoplot(bw_lm2, which = 3, ncol = 1) |> print()
```

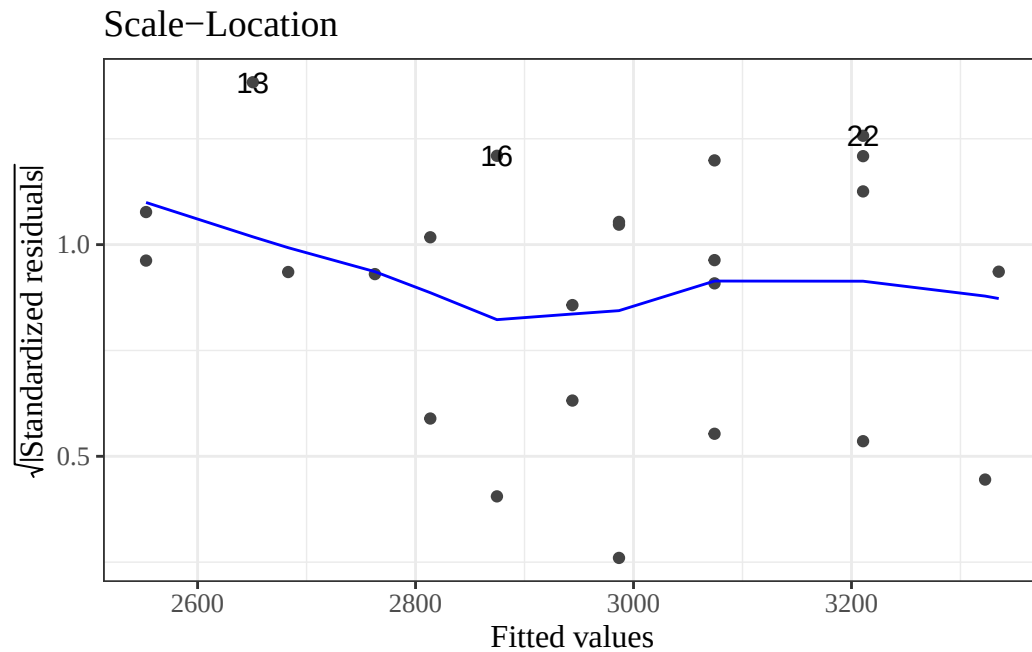


Figure 29: Scale-location plot of `birthweight` data

Here, the blue line doesn't need to be near 0, but it should be flat. If not, the residual variance σ^2 might not be constant, and we might need to transform our outcome Y (or use a model that allows non-constant variance).

7.6.3 Residuals versus leverage

We can also plot our standardized residuals against “leverage”, which roughly speaking is a measure of how unusual each x_i value is. Very unusual x_i values can have extreme effects on the model fit, so we might want to remove those observations as outliers, particularly if they have large residuals.


```
autoplot(bw_lm2, which = 5, ncol = 1) |> print()
```

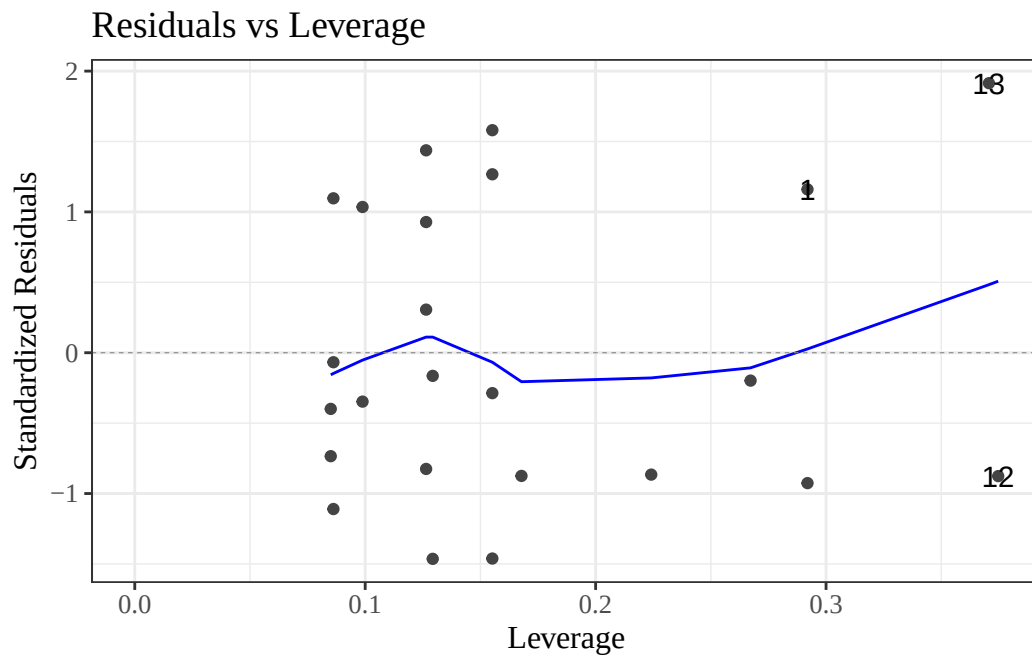


Figure 30: birthweight model with interactions (Equation 2): residuals versus leverage

The blue line should be relatively flat and close to 0 here.

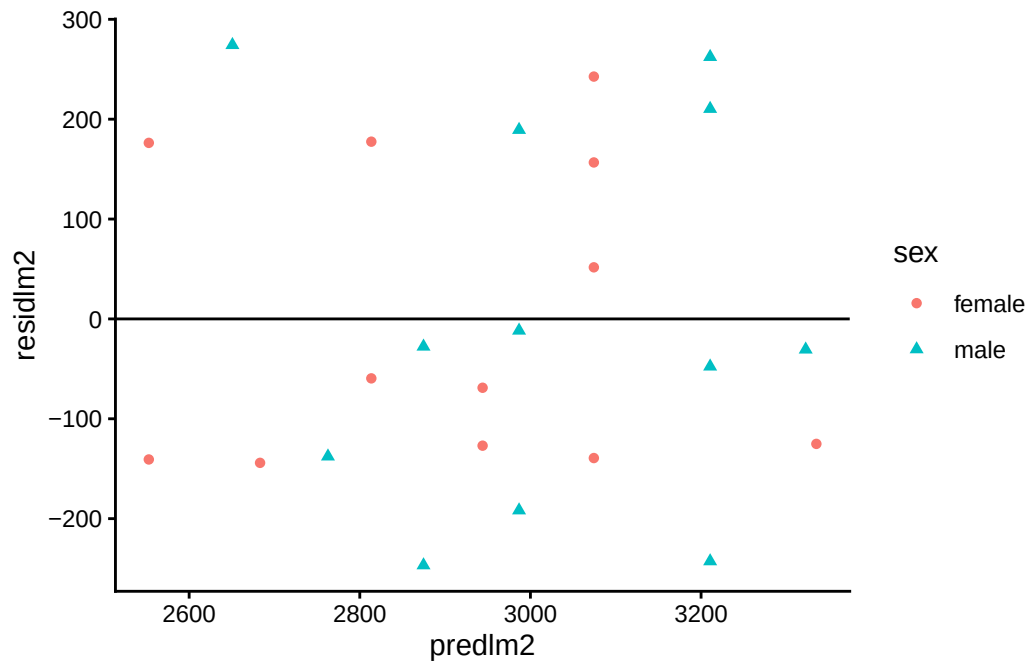
7.7 Diagnostics constructed by hand

```
bw <-
  bw |>
  mutate(
    predlm2 = predict(bw_lm2),
    residlm2 = weight - predlm2,
    std_resid = residlm2 / sigma(bw_lm2),
    # std_resid_builtin = rstandard(bw_lm2), # uses leverage
    sqrt_abs_std_resid = std_resid |> abs() |> sqrt()
  )
```

Residuals vs fitted

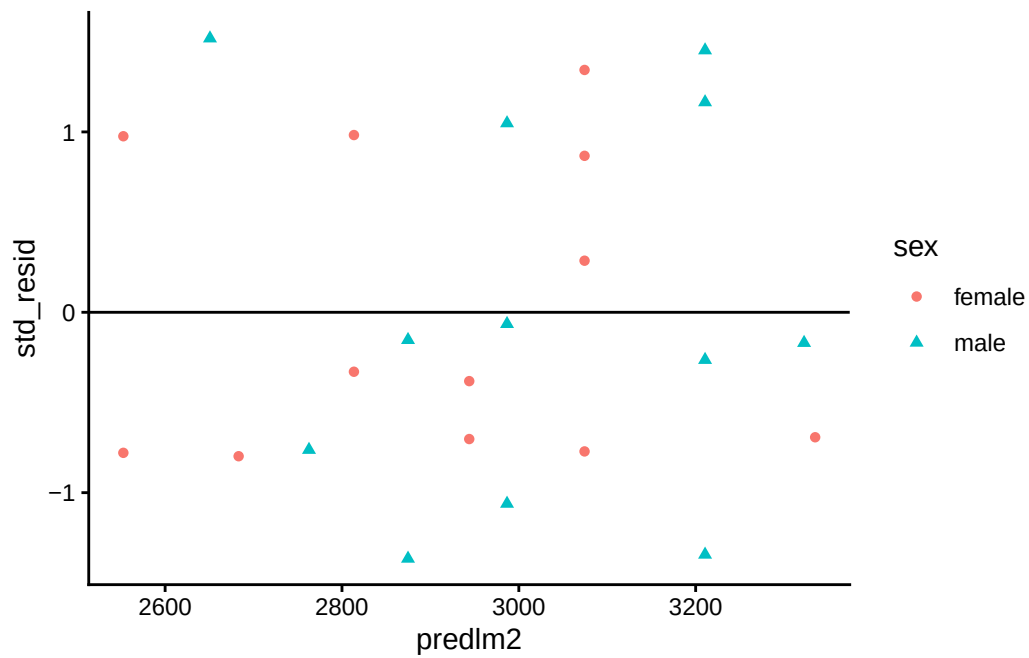
```
resid_vs_fit <- bw |>
  ggplot(
    aes(x = predlm2, y = residlm2, col = sex, shape = sex)
  ) +
  geom_point() +
  theme_classic() +
  geom_hline(yintercept = 0)

print(resid_vs_fit)
```



Standardized residuals vs fitted

```
bw |>
  ggplot(
    aes(x = predlm2, y = std_resid, col = sex, shape = sex)
  ) +
  geom_point() +
  theme_classic() +
  geom_hline(yintercept = 0)
```



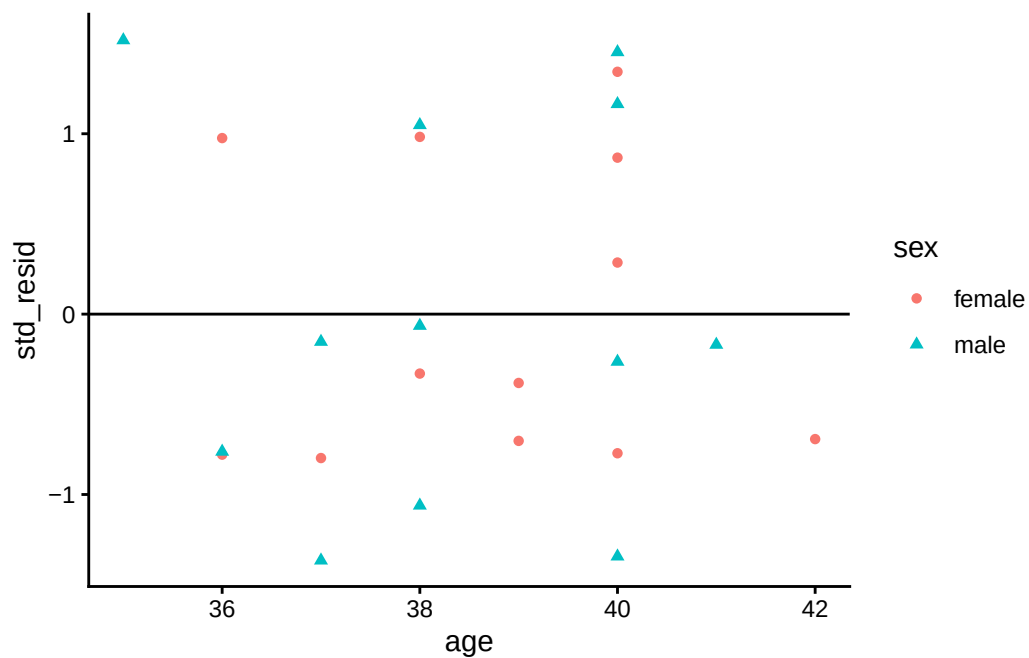
Standardized residuals vs gestational age

```
bw |>
  ggplot(
```

```

  aes(x = age, y = std_resid, col = sex, shape = sex)
) +
geom_point() +
theme_classic() +
geom_hline(yintercept = 0)

```



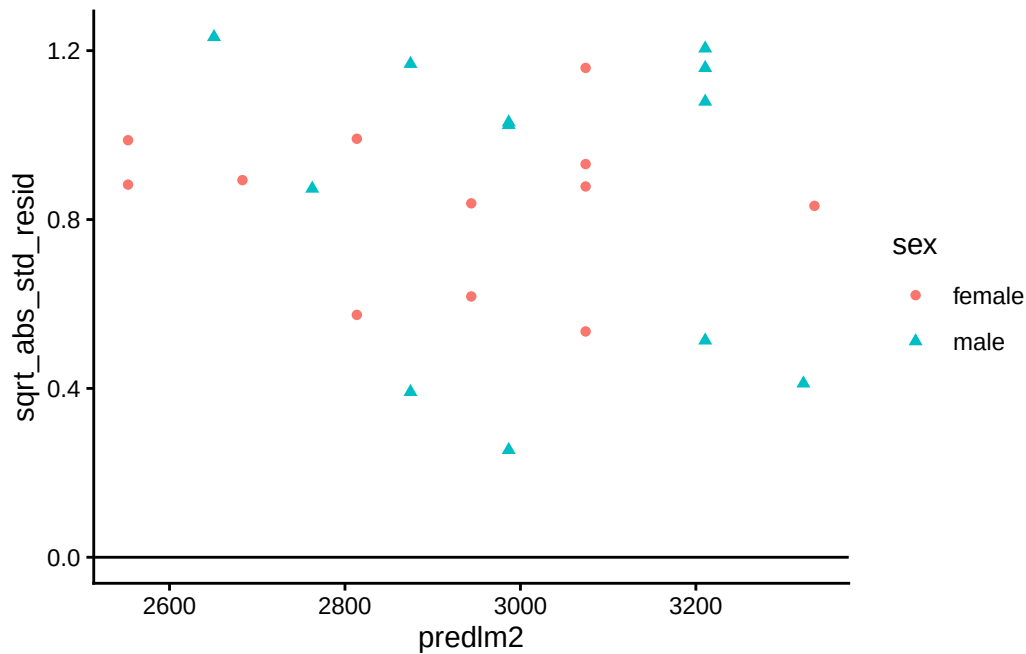
`sqrt(abs(rstandard()))` vs fitted

Compare with `autoplot(bw_lm2, 3)`

```

bw |>
  ggplot(
    aes(x = predlm2, y = sqrt_abs_std_resid, col = sex, shape = sex)
  ) +
  geom_point() +
  theme_classic() +
  geom_hline(yintercept = 0)

```



8 Model selection

(adapted from Dobson and Barnett (2018) §6.3.3; for more information on prediction, see James et al. (2013) and Harrell (2015)).

If we have a lot of covariates in our dataset, we might want to choose a small subset to use in our model.

There are a few possible metrics to consider for choosing a “best” model.

8.1 Mean squared error

We might want to minimize the **mean squared error**, $E[(y - \hat{y})^2]$, for new observations that weren’t in our data set when we fit the model.

Unfortunately,

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

gives a biased estimate of $E[(y - \hat{y})^2]$ for new data. If we want an unbiased estimate, we will have to be clever.

8.1.1 Cross-validation

```
data("carbohydrate", package = "dobson")
library(cvTools)
full_model <- lm(carbohydrate ~ ., data = carbohydrate)
cv_full <-
  full_model |> cvFit(
    data = carbohydrate, K = 5, R = 10,
    y = carbohydrate$carbohydrate
  )

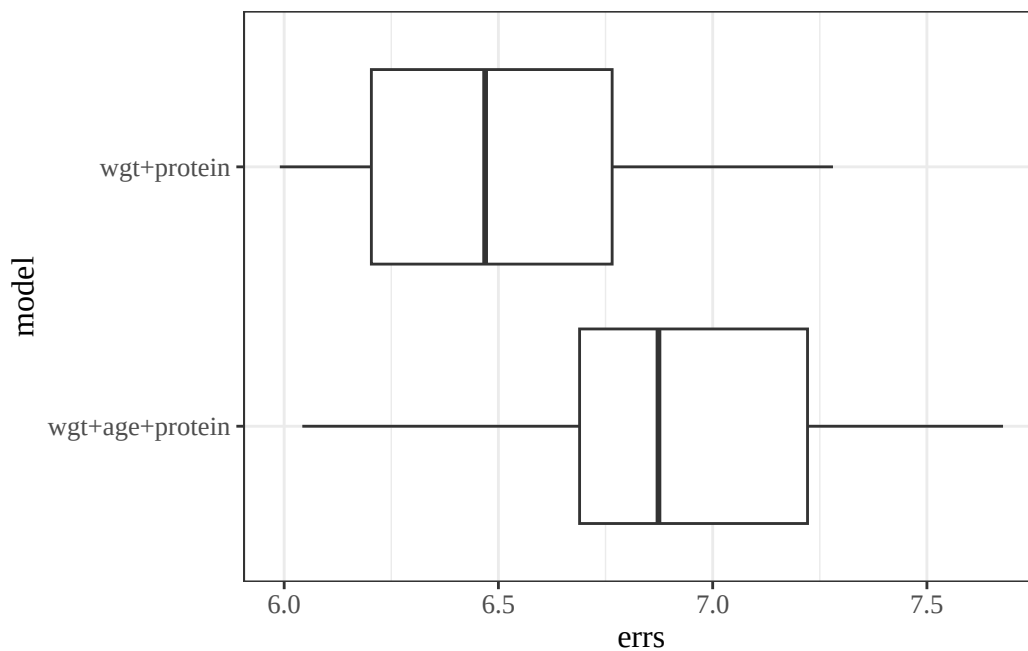
reduced_model <- full_model |> update(formula = ~ . - age)
```

```
cv_reduced <-
  reduced_model |> cvFit(
    data = carbohydrate, K = 5, R = 10,
    y = carbohydrate$carbohydrate
  )
```

```
results_reduced <-
  tibble(
    model = "wgt+protein",
    errs = cv_reduced$reps[]
  )
results_full <-
  tibble(
    model = "wgt+age+protein",
    errs = cv_full$reps[]
  )

cv_results <-
  bind_rows(results_reduced, results_full)

cv_results |>
  ggplot(aes(y = model, x = errs)) +
  geom_boxplot()
```



comparing metrics

```
compare_results <- tribble(
  ~model, ~cvRMSE, ~r.squared, ~adj.r.squared, ~trainRMSE, ~loglik,
  "full",
  cv_full$cv,
  summary(full_model)$r.squared,
  summary(full_model)$adj.r.squared,
  sigma(full_model),
```

```

logLik(full_model) |> as.numeric(),
"reduced",
cv_reduced$cv,
summary(reduced_model)$r.squared,
summary(reduced_model)$adj.r.squared,
sigma(reduced_model),
logLik(reduced_model) |> as.numeric()
)

compare_results
#> # A tibble: 2 x 6
#>   model    cvRMSE r.squared adj.r.squared trainRMSE loglik
#>   <chr>    <dbl>    <dbl>         <dbl>    <dbl>  <dbl>
#> 1 full      6.92    0.481         0.383     5.96  -61.8
#> 2 reduced   6.53    0.445         0.380     5.97  -62.5

```

```

anova(full_model, reduced_model)
#> # A tibble: 2 x 6
#>   Res.Df  RSS    Df `Sum of Sq`    F `Pr(>F)`
#>   <dbl> <dbl> <dbl>    <dbl> <dbl>    <dbl>
#> 1     16  568.   NA      NA    NA      NA
#> 2     17  606.  -1    -38.4  1.08    0.314

```

8.1.2 stepwise regression

Caution about stepwise selection

Stepwise regression has several known problems:

- It tends to select too many variables (overfitting)
- P-values and confidence intervals are biased after selection
- It ignores model uncertainty
- Results can be unstable across different samples

Consider using cross-validation, penalized methods (like Lasso), or subject-matter knowledge instead. See Harrell (2015) and Heinze et al. (2018) for more discussion.

```

library(olsrr)
olsrr::ols_step_both_aic(full_model)
#>
#>
#>                               Stepwise Summary
#> -----
#> Step    Variable      AIC      SBC      SBIC      R2      Adj. R2
#> -----
#> 0      Base Model    140.773    142.764    83.068    0.00000    0.00000
#> 1      protein (+)   137.950    140.937    80.438    0.21427    0.17061
#> 2      weight (+)    132.981    136.964    77.191    0.44544    0.38020
#> -----
#>
#> Final Model Output
#> -----
#>
#>                               Model Summary
#> -----
#> R                               0.667      RMSE                               5.505

```

```

#> R-Squared          0.445      MSE          30.301
#> Adj. R-Squared     0.380      Coef. Var     15.879
#> Pred R-Squared     0.236      AIC          132.981
#> MAE                4.593      SBC          136.964
#> -----
#> RMSE: Root Mean Square Error
#> MSE: Mean Square Error
#> MAE: Mean Absolute Error
#> AIC: Akaike Information Criteria
#> SBC: Schwarz Bayesian Criteria
#>
#>
#> ANOVA
#> -----
#>
#> Sum of
#> Squares      DF      Mean Square      F      Sig.
#> -----
#> Regression    486.778      2      243.389      6.827      0.0067
#> Residual      606.022     17      35.648
#> Total        1092.800     19
#> -----
#>
#>
#> Parameter Estimates
#> -----
#>
#> model      Beta      Std. Error      Std. Beta      t      Sig.      lower      upper
#> -----
#> (Intercept) 33.130      12.572           2.635      2.635      0.017      6.607      59.654
#> protein     1.824       0.623           0.534      2.927      0.009      0.509      3.139
#> weight     -0.222       0.083          -0.486     -2.662      0.016     -0.397     -0.046
#> -----

```

8.1.3 Lasso

$$\arg \max_{\theta} \left\{ \ell(\theta) - \lambda \sum_{j=1}^p |\beta_j| \right\}$$

```

library(glmnet)
y <- carbohydrate$carbohydrate
x <- carbohydrate |>
  select(age, weight, protein) |>
  as.matrix()
fit <- glmnet(x, y)

```

```

autoplot(fit, xvar = "lambda")

```

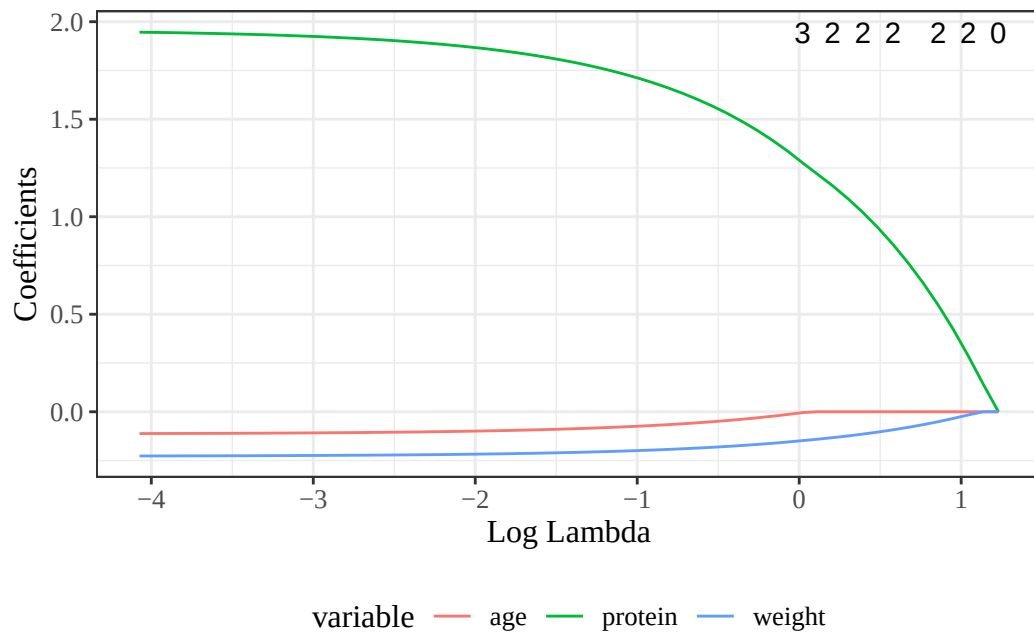
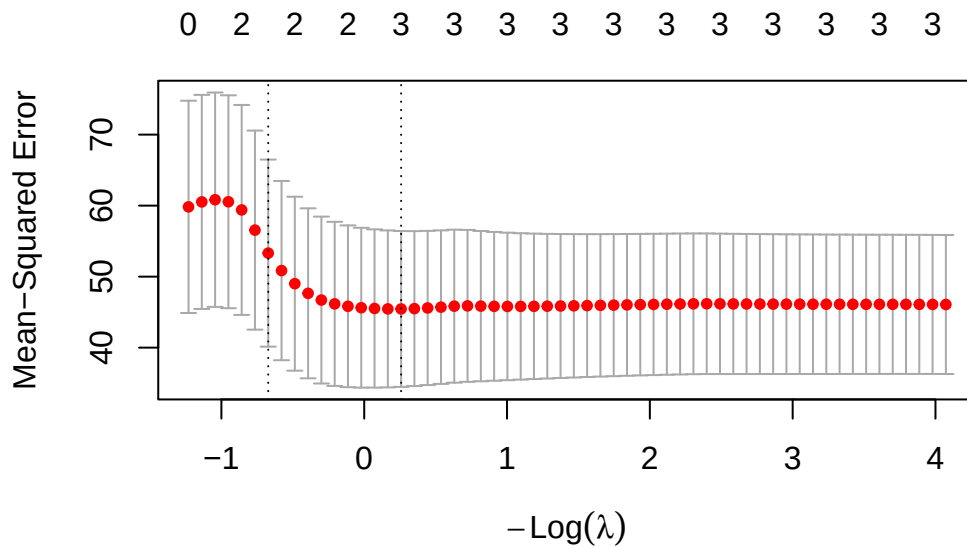


Figure 31: Lasso selection

```
cvfit <- cv.glmnet(x, y)
plot(cvfit)
```



```
coef(cvfit, s = "lambda.1se")
#> 4 x 1 sparse Matrix of class "dgCMatrix"
#>      lambda.1se
#> (Intercept) 34.3091615
#> age         .
#> weight      -0.0800142
```


Table 20: The `iris` data

```
head(iris)
#> # A tibble: 6 x 5
#>   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
#>   <dbl>         <dbl>         <dbl>         <dbl> <fct>
#> 1         5.1         3.5         1.4         0.2 setosa
#> 2         4.9         3         1.4         0.2 setosa
#> 3         4.7         3.2         1.3         0.2 setosa
#> 4         4.6         3.1         1.5         0.2 setosa
#> 5         5         3.6         1.4         0.2 setosa
#> 6         5.4         3.9         1.7         0.4 setosa
```

```
#> protein      0.7640510
```

9 Categorical covariates with more than two levels

9.1 Example: birthweight

In the birthweight example, the variable `sex` had only two observed values:

```
unique(bw$sex)
#> [1] female male
#> Levels: female male
```

If there are more than two observed values, we can't just use a single variable with 0s and 1s.

9.2

For example, Table 20 shows the (in)famous¹² `iris` data (Anderson (1935)), and Table 21 provides summary statistics. The data include three species: “setosa”, “versicolor”, and “virginica”.

```
library(table1)
table1(
  x = ~ . | Species,
  data = iris,
  overall = FALSE
)
```

If we want to model `Sepal.Length` by species, we could create a variable X that represents “setosa” as $X = 1$, “virginica” as $X = 2$, and “versicolor” as $X = 3$.

Then we could fit a model like:

```
iris_lm1 <- lm(Sepal.Length ~ X, data = iris)
iris_lm1 |>
  parameters() |>
  print_md()
```

Table 23: Model of `iris` data with numeric coding of `Species`

Parameter	Coefficient	SE	95% CI	t(148)	p
(Intercept)	4.91	0.16	(4.60, 5.23)	30.83	< .001
X	0.46	0.07	(0.32, 0.61)	6.30	< .001

¹²<https://www.meganstodel.com/posts/no-to-iris/>

Table 21: Summary statistics for the `iris` data

	setosa (N=50)	versicolor (N=50)	virginica (N=50)
Sepal.Length			
Mean (SD)	5.01 (0.352)	5.94 (0.516)	6.59 (0.636)
Median [Min, Max]	5.00 [4.30, 5.80]	5.90 [4.90, 7.00]	6.50 [4.90, 7.90]
Sepal.Width			
Mean (SD)	3.43 (0.379)	2.77 (0.314)	2.97 (0.322)
Median [Min, Max]	3.40 [2.30, 4.40]	2.80 [2.00, 3.40]	3.00 [2.20, 3.80]
Petal.Length			
Mean (SD)	1.46 (0.174)	4.26 (0.470)	5.55 (0.552)
Median [Min, Max]	1.50 [1.00, 1.90]	4.35 [3.00, 5.10]	5.55 [4.50, 6.90]
Petal.Width			
Mean (SD)	0.246 (0.105)	1.33 (0.198)	2.03 (0.275)
Median [Min, Max]	0.200 [0.100, 0.600]	1.30 [1.00, 1.80]	2.00 [1.40, 2.50]

Table 22: `iris` data with numeric coding of species

```

data(iris) # this step is not always necessary, but ensures you're starting
# from the original version of a dataset stored in a loaded package

iris <-
  iris |>
  tibble() |>
  mutate(
    X = case_when(
      Species == "setosa" ~ 1,
      Species == "virginica" ~ 2,
      Species == "versicolor" ~ 3
    )
  )

iris |>
  distinct(Species, X)
#> # A tibble: 3 x 2
#>   Species      X
#>   <fct>    <dbl>
#> 1 setosa      1
#> 2 versicolor  3
#> 3 virginica   2

```

9.3 Let's see how that model looks:

```
iris_plot1 <- iris |>
  ggplot(
    aes(
      x = X,
      y = Sepal.Length
    )
  ) +
  geom_point(alpha = .1) +
  geom_abline(
    intercept = coef(iris_lm1)[1],
    slope = coef(iris_lm1)[2]
  ) +
  theme_bw(base_size = 18)
print(iris_plot1)
```

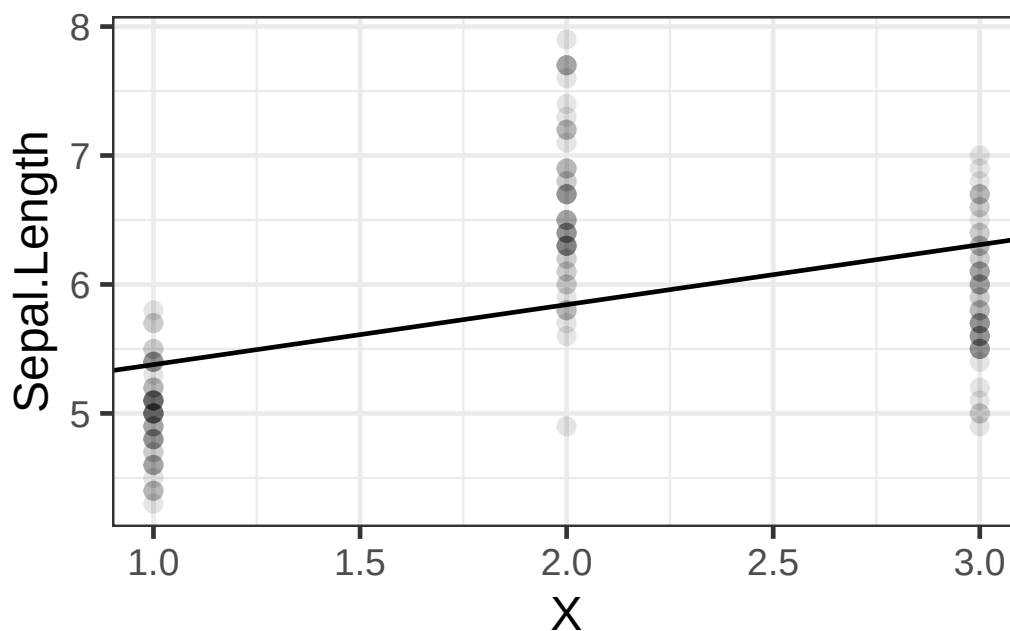


Figure 32: Model of `iris` data with numeric coding of `Species`

We have forced the model to use a straight line for the three estimated means. Maybe not a good idea?

9.4 Let's see what R does with categorical variables by default:

```
iris_lm2 <- lm(Sepal.Length ~ Species, data = iris)
iris_lm2 |>
  parameters() |>
  print_md()
```

Table 24: Model of `iris` data with `Species` as a categorical variable

Parameter	Coefficient	SE	95% CI	t(147)	p
(Intercept)	5.01	0.07	(4.86, 5.15)	68.76	< .001
Species (versicolor)	0.93	0.10	(0.73, 1.13)	9.03	< .001

Table 24: Model of `iris` data with `Species` as a categorical variable

Parameter	Coefficient	SE	95% CI	t(147)	p
Species (virginica)	1.58	0.10	(1.38, 1.79)	15.37	< .001

9.5 Re-parametrize with no intercept

If you don't want the default and offset option, you can use “-1” like we've seen previously:

```
iris_lm2_no_int <- lm(Sepal.Length ~ Species - 1, data = iris)
iris_lm2_no_int |>
  parameters() |>
  print_md()
```

Table 25

Parameter	Coefficient	SE	95% CI	t(147)	p
Species (setosa)	5.01	0.07	(4.86, 5.15)	68.76	< .001
Species (versicolor)	5.94	0.07	(5.79, 6.08)	81.54	< .001
Species (virginica)	6.59	0.07	(6.44, 6.73)	90.49	< .001

9.6 Let's see what these new models look like:

```
iris_plot2 <-
  iris |>
  mutate(
    predlm2 = predict(iris_lm2)
  ) |>
  arrange(X) |>
  ggplot(aes(x = X, y = Sepal.Length)) +
  geom_point(alpha = .1) +
  geom_line(aes(y = predlm2), col = "red") +
  geom_abline(
    intercept = coef(iris_lm1)[1],
    slope = coef(iris_lm1)[2]
  ) +
  theme_bw(base_size = 18)

print(iris_plot2)
```

```

formula(iris_lm2)
#> Sepal.Length ~ Species
model.matrix(iris_lm2) |>
  as_tibble() |>
  unique()
#> # A tibble: 3 x 3
#>   `(Intercept)` Speciesversicolor Speciesvirginica
#>   <dbl>          <dbl>          <dbl>
#> 1         1         0         0
#> 2         1         1         0
#> 3         1         0         1

```

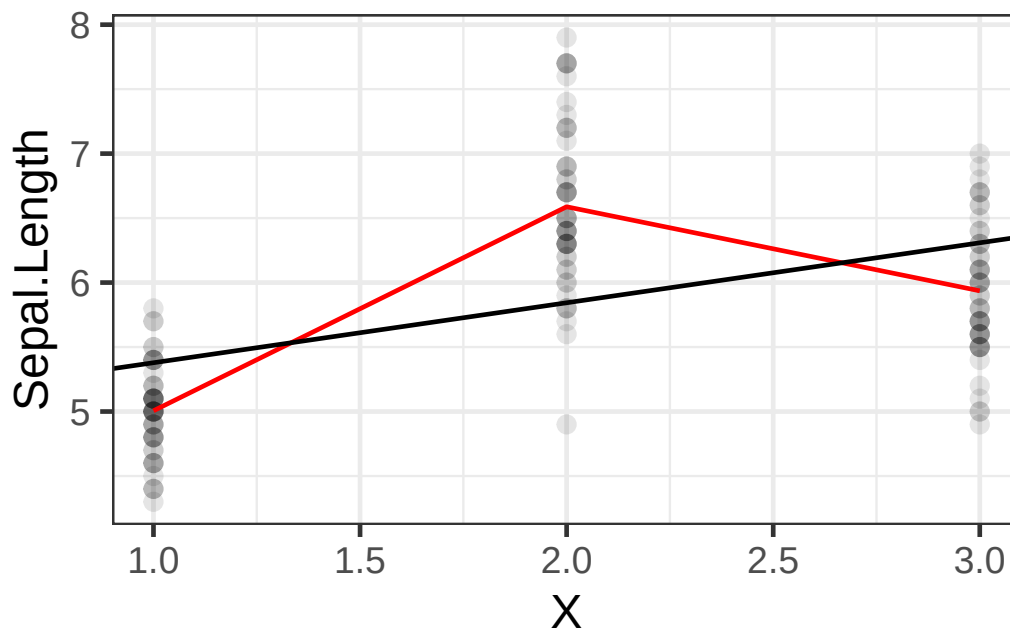


Figure 33

9.7 Let's see how R did that:

This format is called a “corner point parametrization” (e.g., in Dobson and Barnett (2018)) or “treatment coding” (e.g., in Dunn and Smyth (2018)).

The default contrasts are controlled by `options("contrasts")`:

```

options("contrasts")
#> $contrasts
#>      unordered      ordered
#> "contr.treatment" "contr.poly"

```

See `?options` for more details.

This format is called a “group point parametrization” (e.g., in Dobson and Barnett (2018)).

There are more options; see Dobson and Barnett (2018) §6.4.1 and the `codingMatrices` package¹³

¹³<https://CRAN.R-project.org/package=codingMatrices>

Table 27

```
formula(iris_lm2_no_int)
#> Sepal.Length ~ Species - 1
model.matrix(iris_lm2_no_int) |>
  as_tibble() |>
  unique()
#> # A tibble: 3 x 3
#>   Speciessetosa Speciesversicolor Speciesvirginica
#>   <dbl>          <dbl>          <dbl>
#> 1           1           0           0
#> 2           0           1           0
#> 3           0           0           1
```

Table 28: HERS dataset

```
hers |> head()
#> # A tibble: 6 x 37
#>   HT age raceth nonwhite smoking drinkany exercise physact globrat
#>   <dbl+lbl> <dbl> <dbl+lbl> <dbl+lbl> <dbl+lbl> <dbl+lbl> <dbl+lbl> <dbl+lbl> <dbl+lbl>
#> 1 0 [placebo] 70 2 [Afr~ 1 [yes] 0 [no] 0 [no] 0 [no] 5 [muc~ 3 [goo~
#> 2 0 [placebo] 62 2 [Afr~ 1 [yes] 0 [no] 0 [no] 0 [no] 1 [muc~ 3 [goo~
#> 3 1 [hormone t~ 69 1 [Whi~ 0 [no] 0 [no] 0 [no] 0 [no] 3 [abo~ 3 [goo~
#> 4 0 [placebo] 64 1 [Whi~ 0 [no] 1 [yes] 1 [yes] 0 [no] 1 [muc~ 3 [goo~
#> 5 0 [placebo] 65 1 [Whi~ 0 [no] 0 [no] 0 [no] 0 [no] 2 [som~ 3 [goo~
#> 6 1 [hormone t~ 68 2 [Afr~ 1 [yes] 0 [no] 1 [yes] 0 [no] 3 [abo~ 3 [goo~
#> # i 28 more variables: poorfair <dbl+lbl>, medcond <dbl>, htnmeds <dbl+lbl>,
#> # statins <dbl+lbl>, diabetes <dbl+lbl>, dmpills <dbl+lbl>,
#> # insulin <dbl+lbl>, weight <dbl>, BMI <dbl>, waist <dbl>, WHR <dbl>,
#> # glucose <dbl>, weight1 <dbl>, BMI1 <dbl>, waist1 <dbl>, WHR1 <dbl>,
#> # glucose1 <dbl>, tchol <dbl>, LDL <dbl>, HDL <dbl>, TG <dbl>, tchol1 <dbl>,
#> # LDL1 <dbl>, HDL1 <dbl>, TG1 <dbl>, SBP <dbl>, DBP <dbl>, age10 <dbl>
```

vignette¹⁴ (Venables (2023)).

10 Ordinal covariates

(c.f. Dobson and Barnett (2018) §2.4.4)

We can create ordinal variables in R using the `ordered()` function¹⁵.

Example 10.1.

```
url <- paste0(
  "https://regression.ucsf.edu/sites/g/files/tkssra6706/",
  "f/wysiwyg/home/data/hersdata.dta"
)
library(haven)
hers <- read_dta(url)
```

For ordinal variables, we might want to use contrasts that respect the ordering of the categories. The default treatment contrasts don't account for ordering.

¹⁴<https://cran.r-project.org/web/packages/codingMatrices/vignettes/codingMatrices.pdf>

¹⁵or equivalently, `factor(ordered = TRUE)`

i Working with ordinal variables

When working with ordinal covariates in linear models:

1. Use `ordered()` or `factor(ordered = TRUE)` to create the variable
2. Consider using polynomial contrasts (`contr.poly`) which respect ordering
3. Alternatively, treat the ordinal variable as numeric if equal spacing is reasonable
4. Check `?codingMatrices::contr.diff` for additional contrast options

See Dobson and Barnett (2018) §2.4.4 for more details on contrasts for ordinal variables.

- Anderson, Edgar. 1935. “The Irises of the Gaspé Peninsula.” *Bulletin of American Iris Society* 59: 2–5.
- Chatterjee, Samprit, and Ali S Hadi. 2015. *Regression Analysis by Example*. John Wiley & Sons. <https://www.wiley.com/en-us/Regression+Analysis+by+Example%2C+4th+Edition-p-9780470055458>.
- Dobson, Annette J, and Adrian G Barnett. 2018. *An Introduction to Generalized Linear Models*. 4th ed. CRC press. <https://doi.org/10.1201/9781315182780>.
- Dunn, Peter K, and Gordon K Smyth. 2018. *Generalized Linear Models with Examples in R*. Vol. 53. Springer. <https://link.springer.com/book/10.1007/978-1-4419-0118-7>.
- Faraway, Julian J. 2025. *Linear Models with R*. <https://www.routledge.com/Linear-Models-with-R/Faraway/p/book/9781032583983>.
- Harrell, Frank E. 2015. *Regression Modeling Strategies: With Applications to Linear Models, Logistic Regression, and Survival Analysis*. 2nd ed. Springer. <https://doi.org/10.1007/978-3-319-19425-7>.
- Heinze, Georg, Christine Wallisch, and Daniela Dunkler. 2018. “Variable Selection – A Review and Recommendations for the Practicing Statistician.” *Biometrical Journal* 60 (3): 431–49. <https://doi.org/10.1002/bimj.201700067>.
- Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman. 2015. *Probability and Statistical Inference*. Ninth edition. Pearson.
- Hulley, Stephen, Deborah Grady, Trudy Bush, et al. 1998. “Randomized Trial of Estrogen Plus Progestin for Secondary Prevention of Coronary Heart Disease in Postmenopausal Women.” *JAMA : The Journal of the American Medical Association* (Chicago, IL) 280 (7): 605–13.
- James, Gareth, Daniela Witten, Trevor Hastie, Robert Tibshirani, et al. 2013. *An Introduction to Statistical Learning*. Vol. 112. Springer. <https://www.statlearning.com/>.
- Kleinbaum, David G, and Mitchel Klein. 2010. *Logistic Regression: A Self-Learning Text*. 3rd ed. Springer. <https://link.springer.com/book/10.1007/978-1-4419-1742-3>.
- Kleinbaum, David G, and Mitchel Klein. 2012. *Survival Analysis: A Self-Learning Text*. 3rd ed. Springer. <https://link.springer.com/book/10.1007/978-1-4419-6646-9>.
- Kleinbaum, David G, Lawrence L Kupper, Azhar Nizam, K Muller, and ES Rosenberg. 2014. *Applied Regression Analysis and Other Multivariable Methods*. 5th ed. Cengage Learning. <https://www.cengage.com/c/applied-regression-analysis-and-other-multivariable-methods-5e-kleinbaum/9781285051086/>.
- Kutner, Michael H, Christopher J Nachtsheim, John Neter, and William Li. 2005. *Applied Linear Statistical Models*. McGraw-Hill.
- Polin, Richard A, William W Fox, and Steven H Abman. 2011. *Fetal and Neonatal Physiology*. 4th

ed. Elsevier health sciences.

Seber, George AF, and Alan J Lee. 2012. *Linear Regression Analysis*. 2nd ed. John Wiley & Sons. <https://www.wiley.com/en-us/Linear+Regression+Analysis%2C+2nd+Edition-p-9781118274422>.

Venables, Bill. 2023. *codingMatrices: Alternative Factor Coding Matrices for Linear Model Formulae*. Version 0.4.0. <https://CRAN.R-project.org/package=codingMatrices>.

Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.

Weisberg, Sanford. 2005. *Applied Linear Regression*. Vol. 528. John Wiley & Sons.